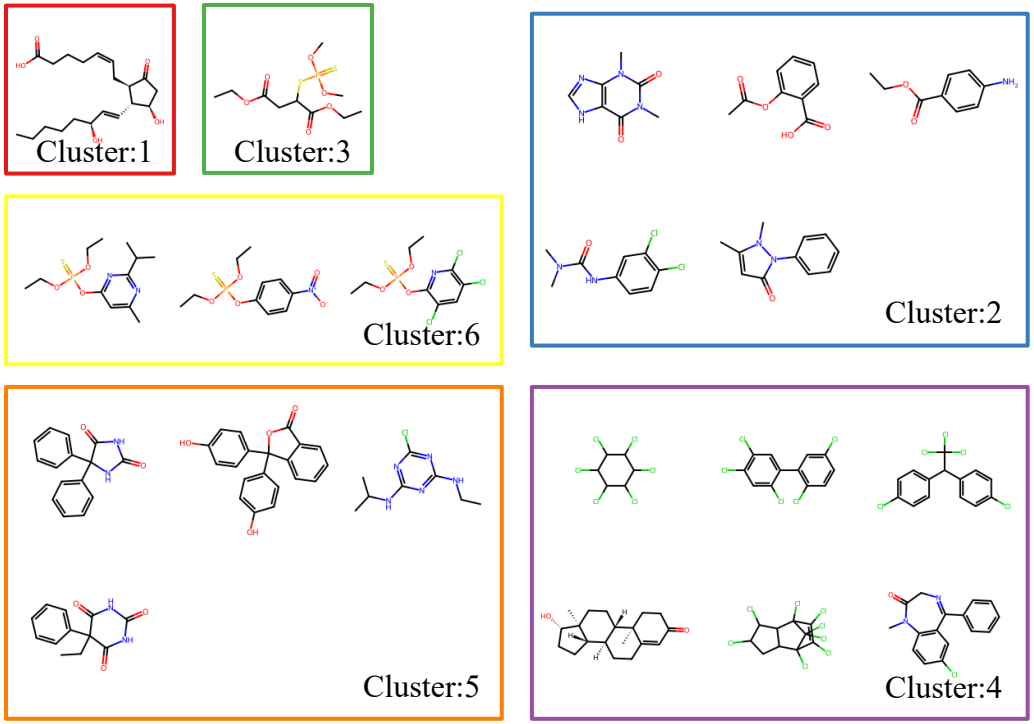
Chapter 3: Uncovering Patterns in Chemistry – Unsupervised Learning and Dimensionality Reduction

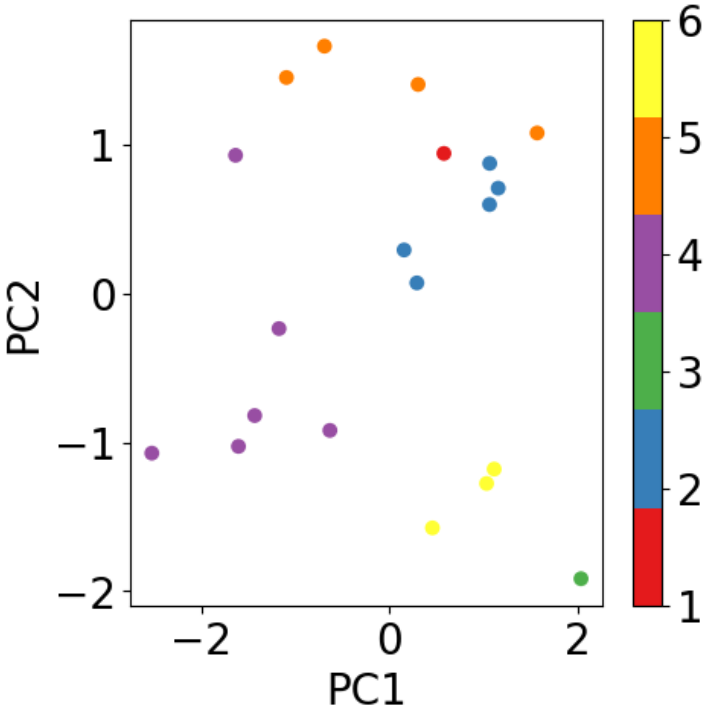
Chem 361

Xuhui Huang

Unsupervised Learning

Find patterns that help better understand data. No labels on data and generally don't make predictions.





Clustering: Group compounds with similar patterns or properties together.

Dimensionality reduction: visualize data in low dimensions (PCs, CVs, reaction coordinates).

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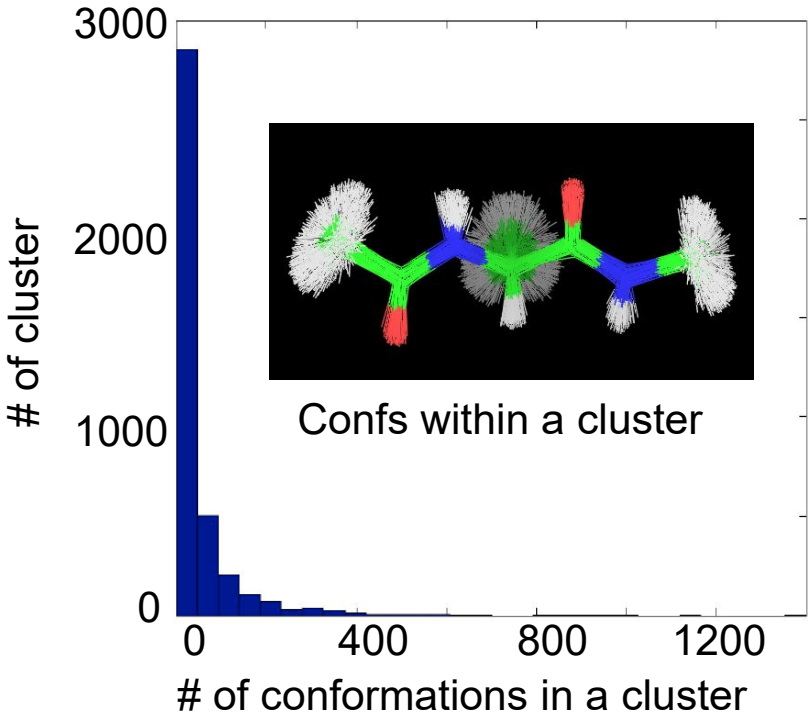
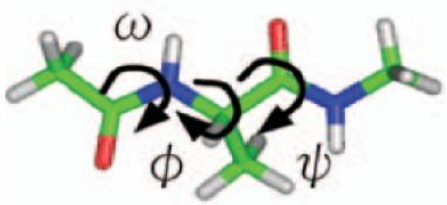
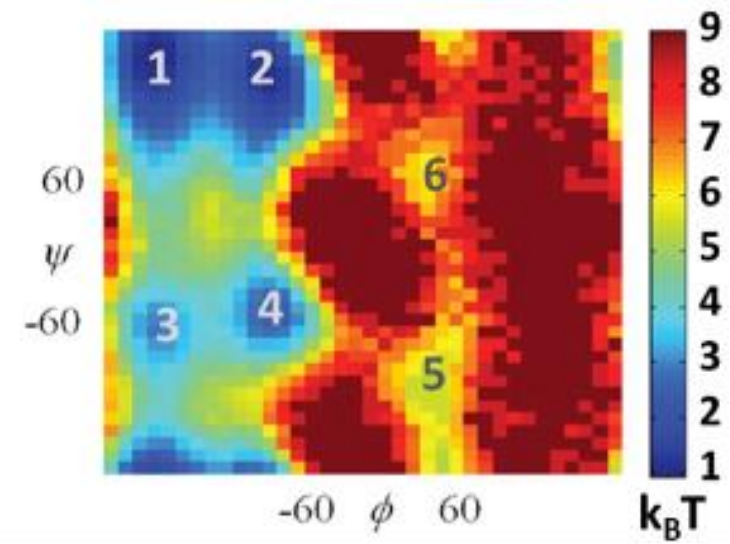
Unsupervised Learning

Find patterns that help better understand data. No labels on data and generally don't make predictions.

Data: 195K conformations of alanine dipeptide







Clustering structurally similar conformations into 3.9K clusters by K-centers.

Dimensionality reduction: visualize data in CVs (2 torsion angles).

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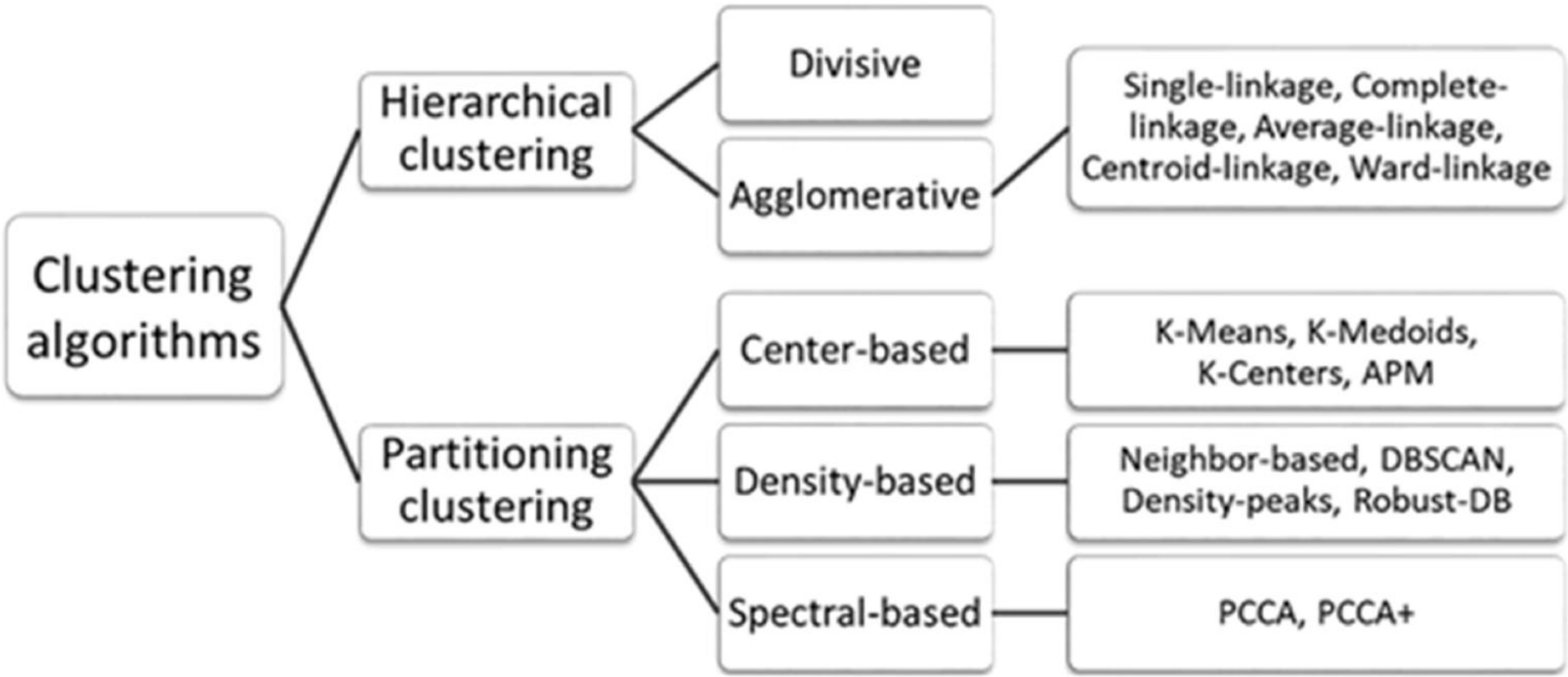
Overview of Clustering Algorithms

Common clustering algorithms for analyzing chemical data:



Chin. J. Chem. Phys. 31, 404 (2018)

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Outline

- **Center-based partitional clustering**
- Density-based partitional clustering
- Hierarchical clustering
- Dimensionality reduction

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K-Means Clustering

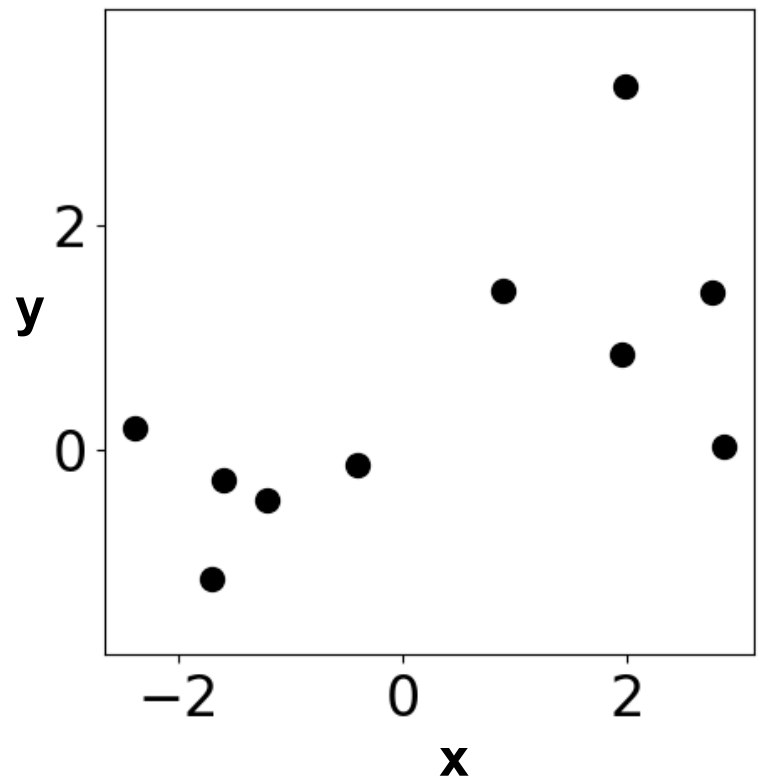
K-means: Partition data points (e.g., chemical feature vectors or structures) into K non-overlapping clusters to minimize the cost function:

$$\min \sum_{i=1}^{K} \sum_{x \in C_i} \|x - \mu_i\|^2$$

C_i denotes to the ith cluster and μ_i is its center (geometric mean).

An iterative procedure (Lloyd's algorithm) is commonly used for its implementation.

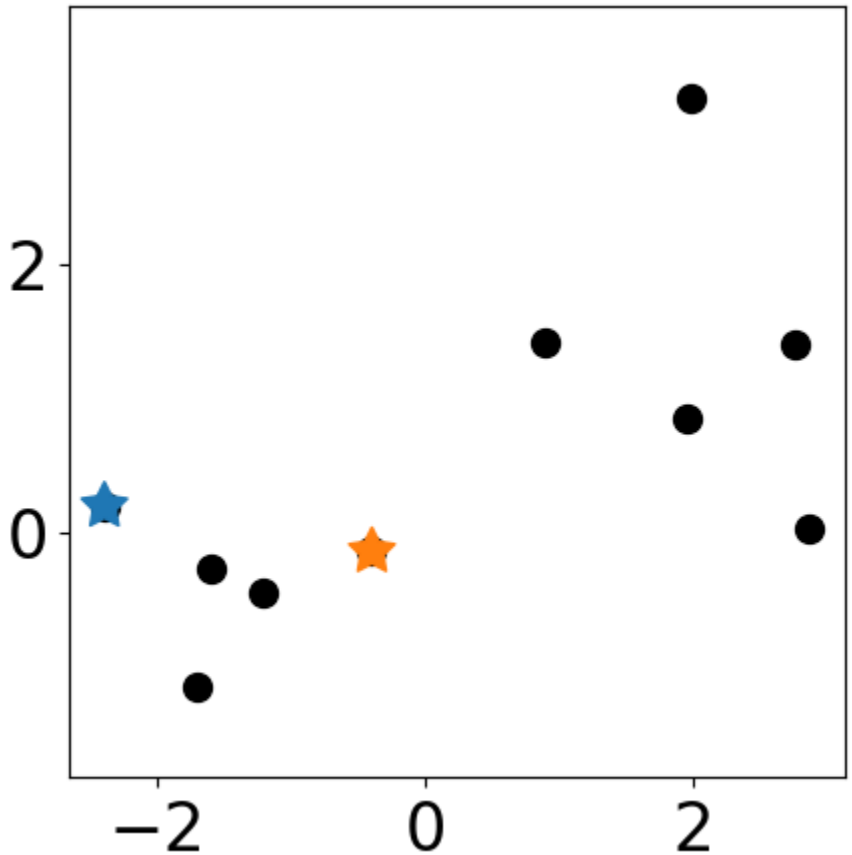
Task: Cluster 10 2D-data points into two clusters (K=2).



S. P. Lloyd, *IEEE Trans. Inf. Theory* 28, 129 (1982)

Lloyd's algorithm for K-Means

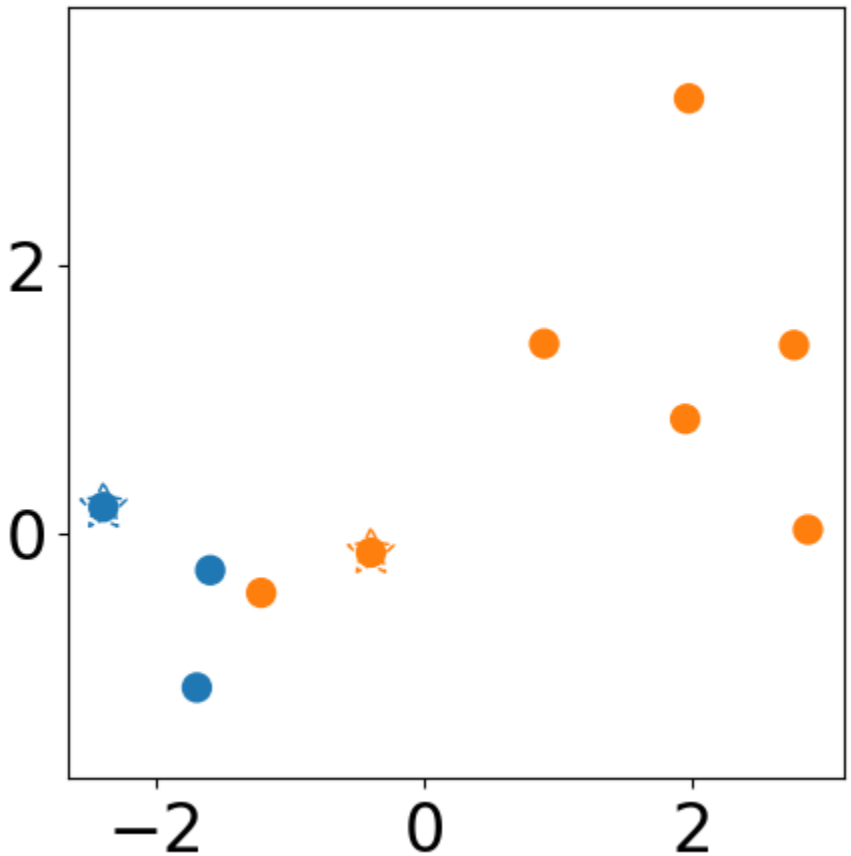
Step 1: Initiation

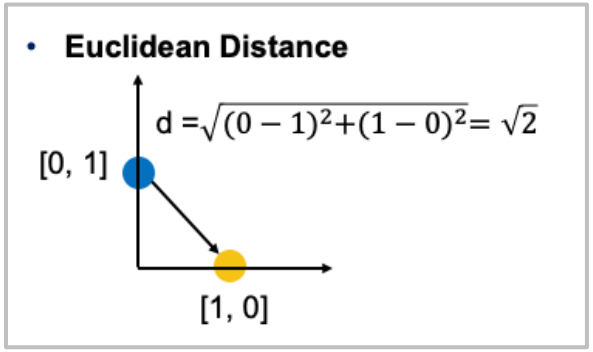


Randomly select two (K=2) data points (stars) as initial cluster centers.

Lloyd's algorithm for K-Means

Step 2: Iteration – assigning data points to centers.

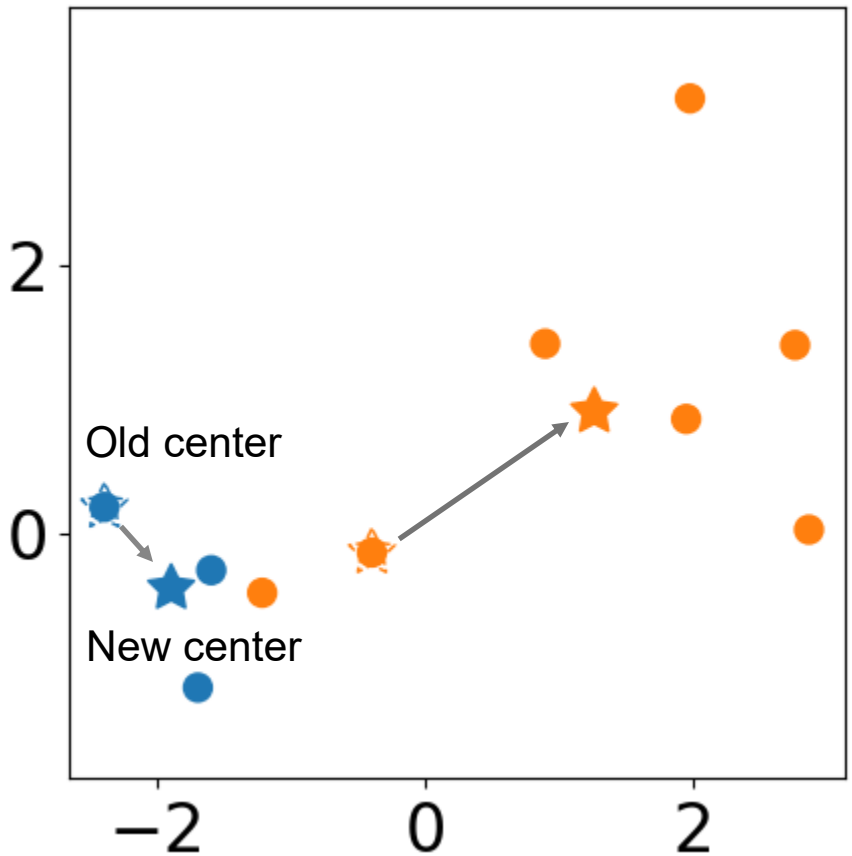




Assign each data point to its nearest center based on Euclidean distance.

Lloyd's algorithm for K-Means

Step 3: Iteration – updating centers

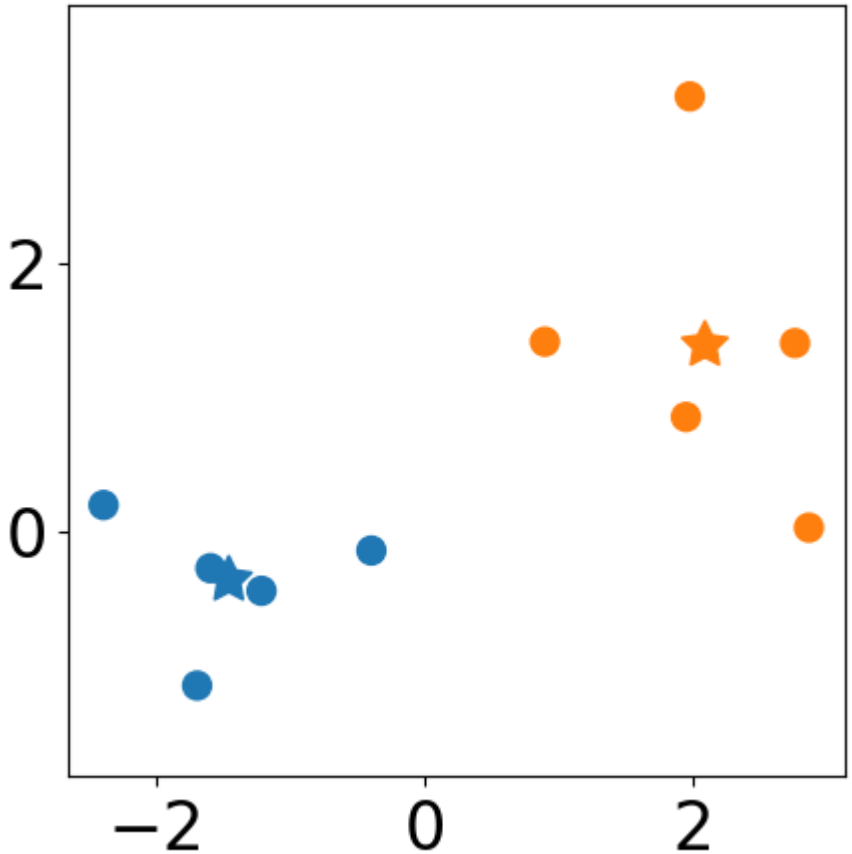


Cluster centers may not locate at existing data points.

Update cluster centers to the new geometric mean of each cluster.

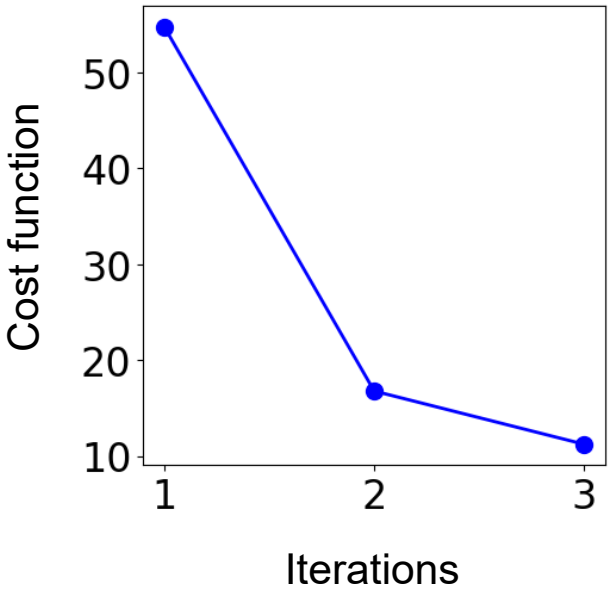
Lloyd's algorithm for K-Means

Step 4: Termination



Cost function to minimize:

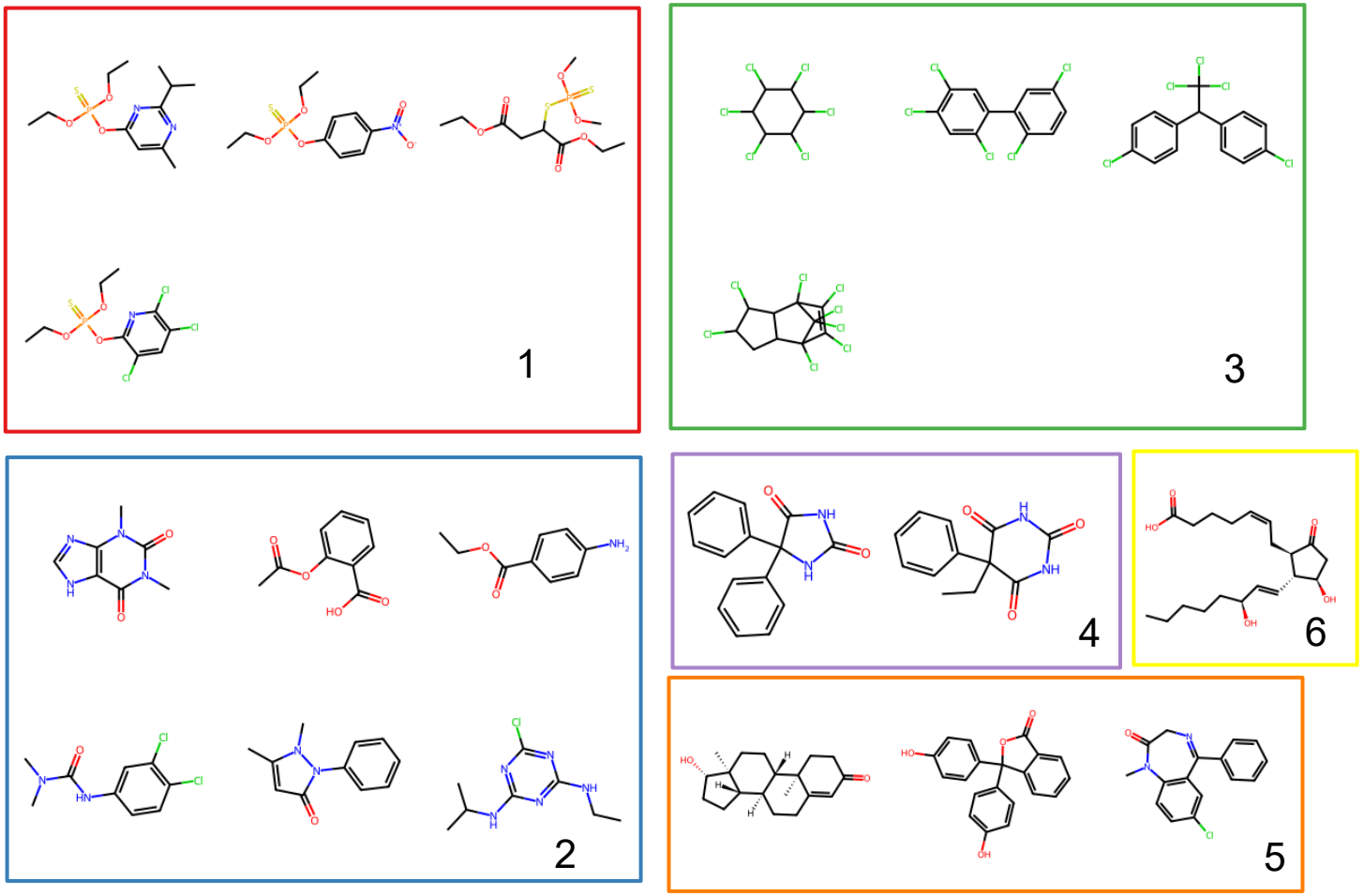
$$\min \sum_{i=1}^{K} \sum_{x \in C_i} \|x - \mu_i\|^2$$



Repeat Steps 2–3 until convergence: either the cost function falls below a threshold, or the maximum iterations are reached – cluster centers no longer change.

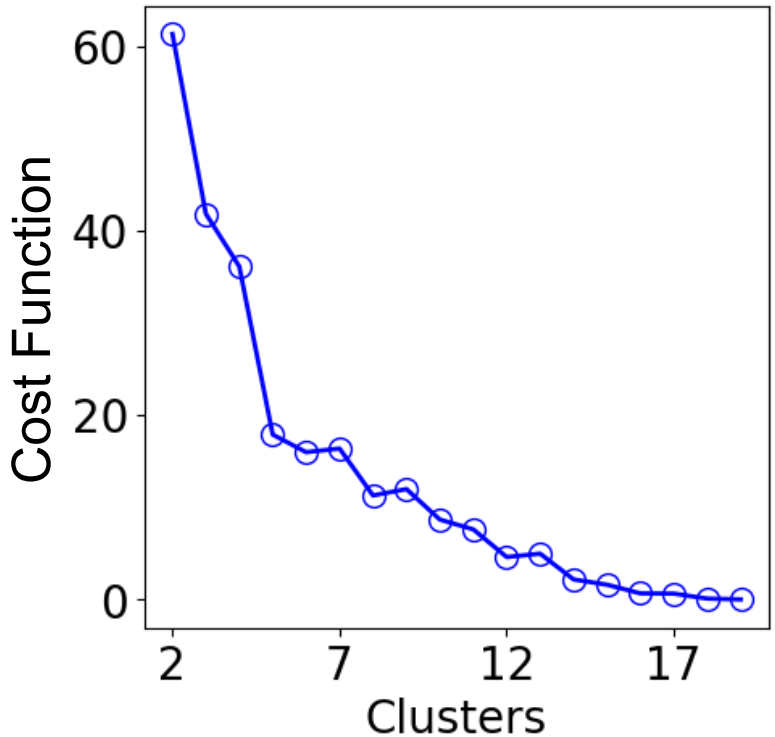
K-means Clustering on Chemical Compounds

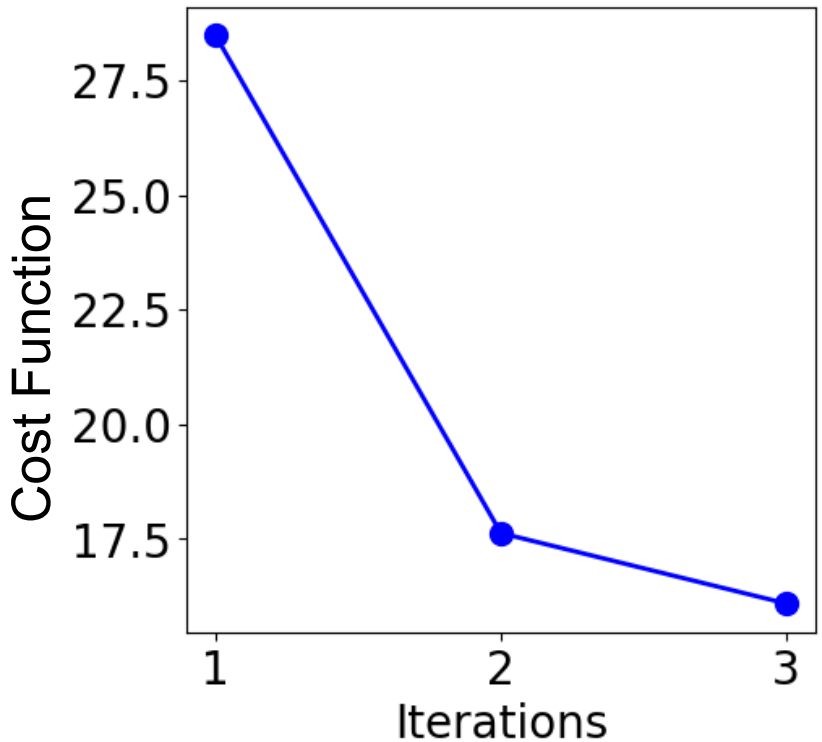
Cluster 20 compounds into 6 groups using Euclidean distance between feature vectors:
[MW, Hdonor, HAcceptor, Rings]



K-means Clustering on Chemical Compounds

Minimize the cost function: $$\min \sum_{i=1}^{K} \sum_{x \in C_i} \|x - \mu_i\|^2$$





We choose K=6.

K-Means Clustering

- K-means minimizes the average distance of data points to their centers – a popular algorithm in chemistry.
- Initial cluster centers affect convergence speed, and K-means may get trapped in local optima.
- The number of clusters (*K*) needs to be determined in prior.
- Its runtime complexity is *O(Knl),* where *n* is the number of data points and *l* is the number of iterations. K-means is slow to converge for large *K* and *n*.
- K-medoids is a variation of K-means where cluster centers must be **actual data points.**

$$\min \sum_{i=1}^{K} \sum_{x \in C_i} \|x - \mathbf{C}(C_i)\|^2$$

C*(Ci)* is cluster center for cluster ***i*.**

K-Centers Clustering

K-centers has a difference cost function than K-means:

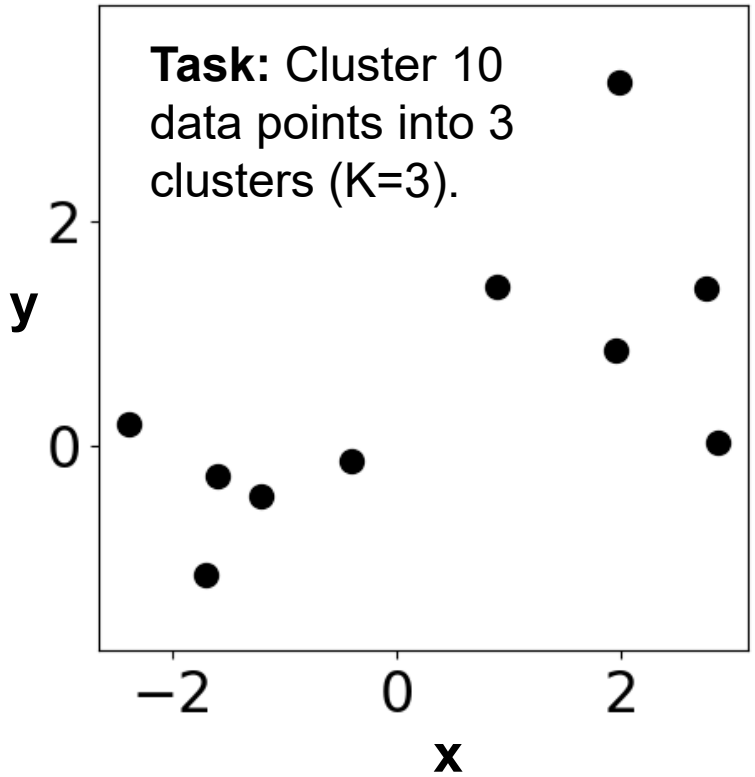
$$\min_i \max_{x \in C_i} d(x, \mu_i)$$

$d(x, \mu_i)$ is the distance between a data point x to its center μ_i.

K-centers algorithm minimizes the maximum distance of each point to all possible centers.

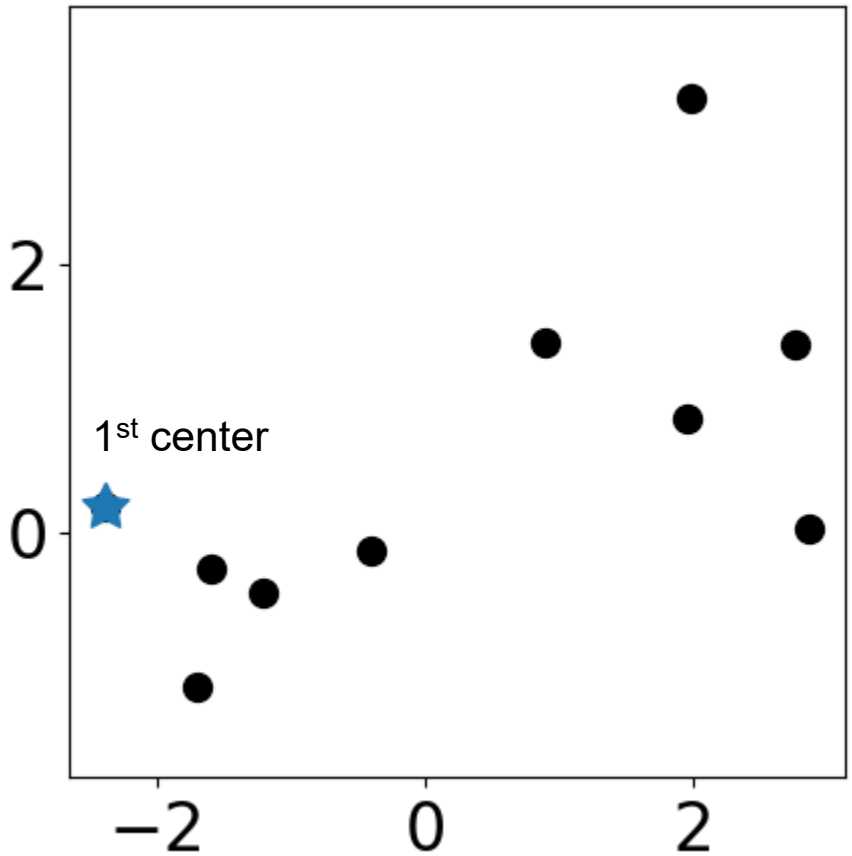
An efficient implementation of K-centers exists for chemical datasets.

J. Comp. Chem., 34, 95-104, (2013)



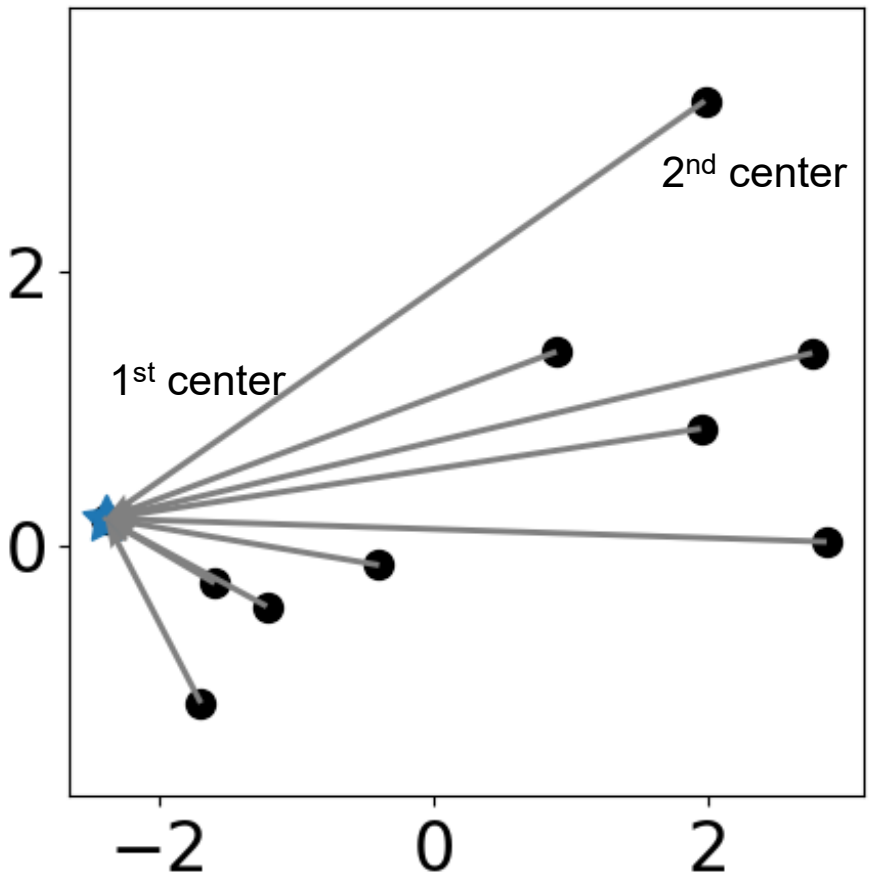
K-Centers Clustering

Step 1: Randomly select the 1st center.



K-Centers Clustering

Step 2: Select the 2nd center as the point furthest from the 1st center.



K-Centers Clustering

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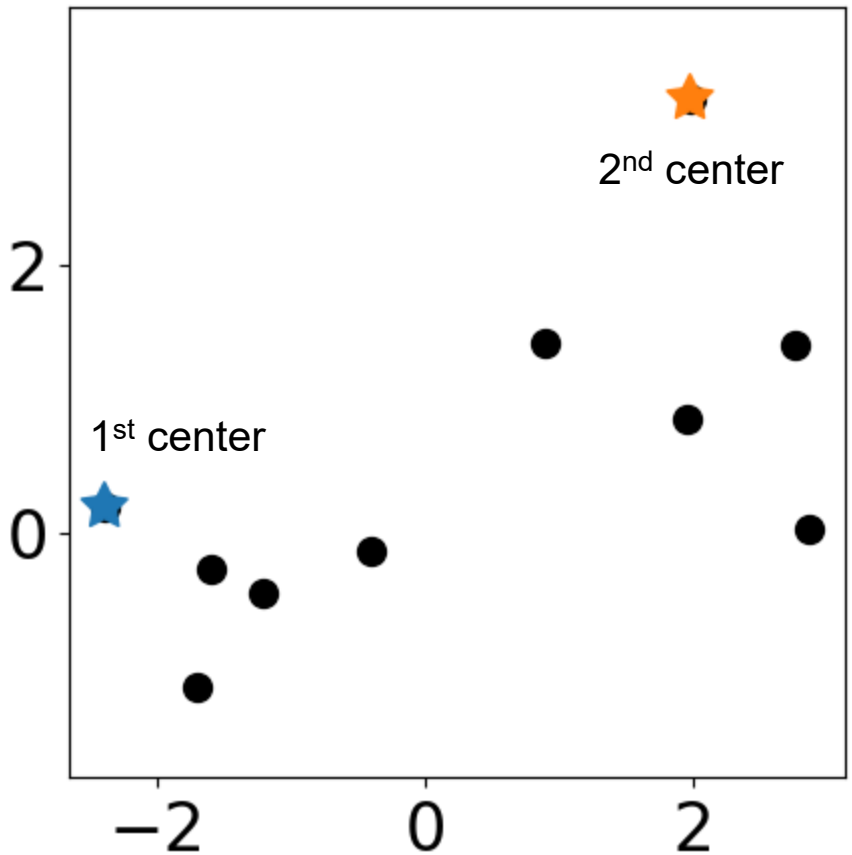
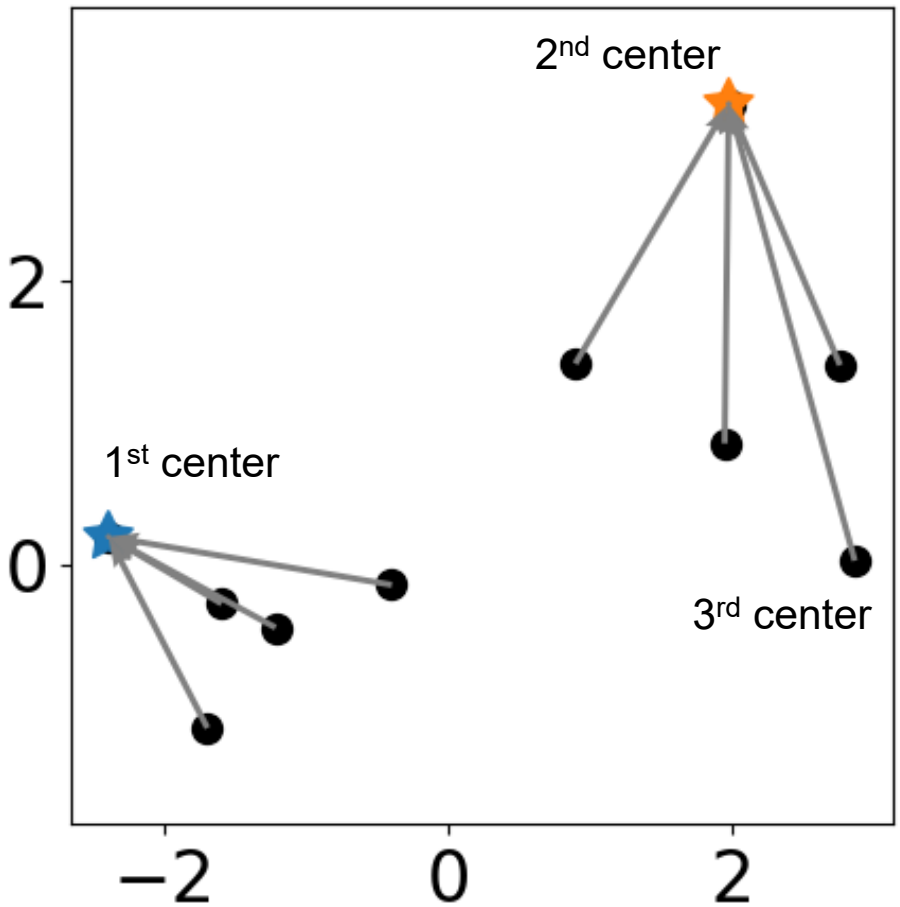


K-Centers Clustering

Step 3: Select the 3rd center as the point furthest from the 1st & 2nd centers.



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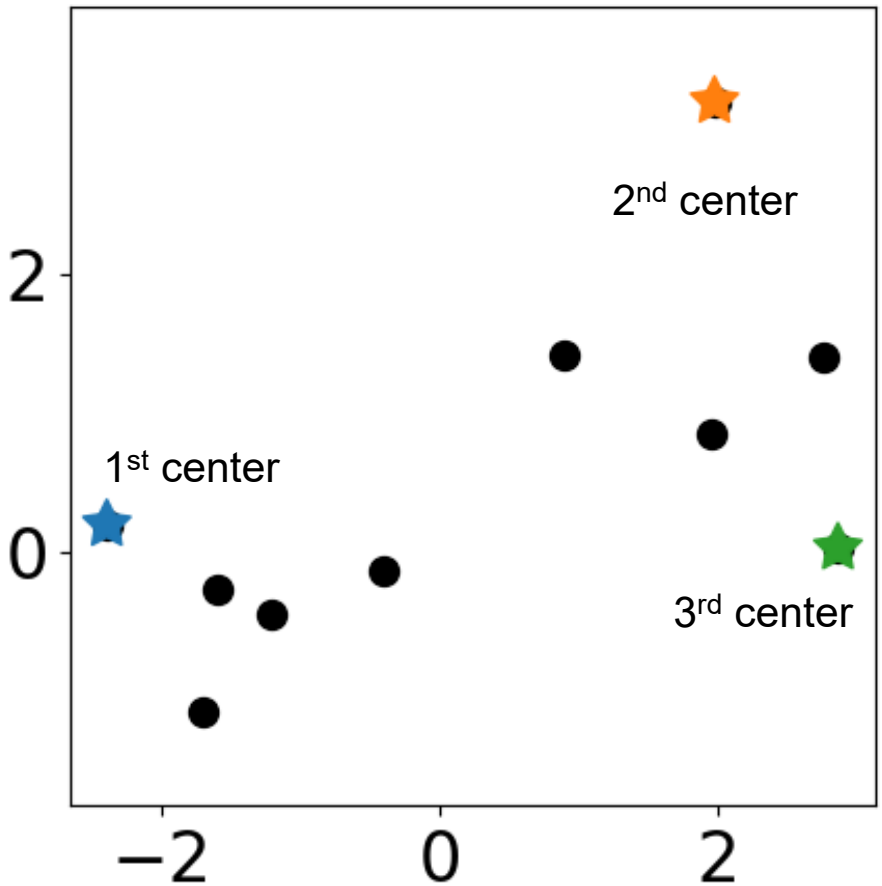

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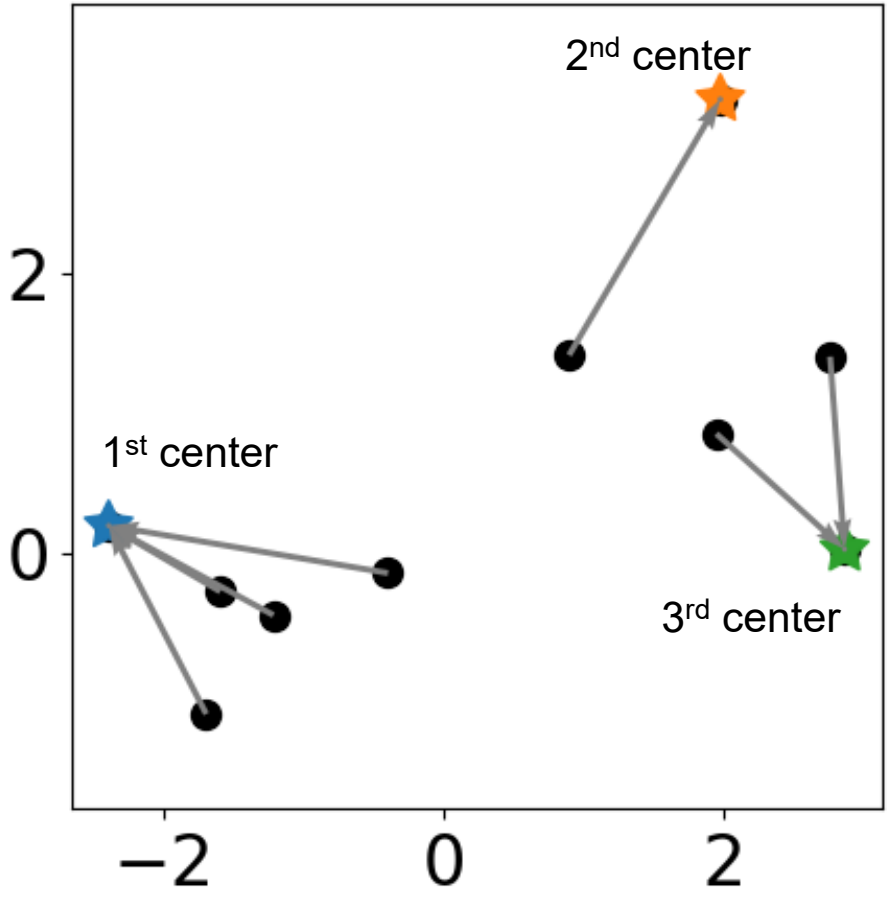
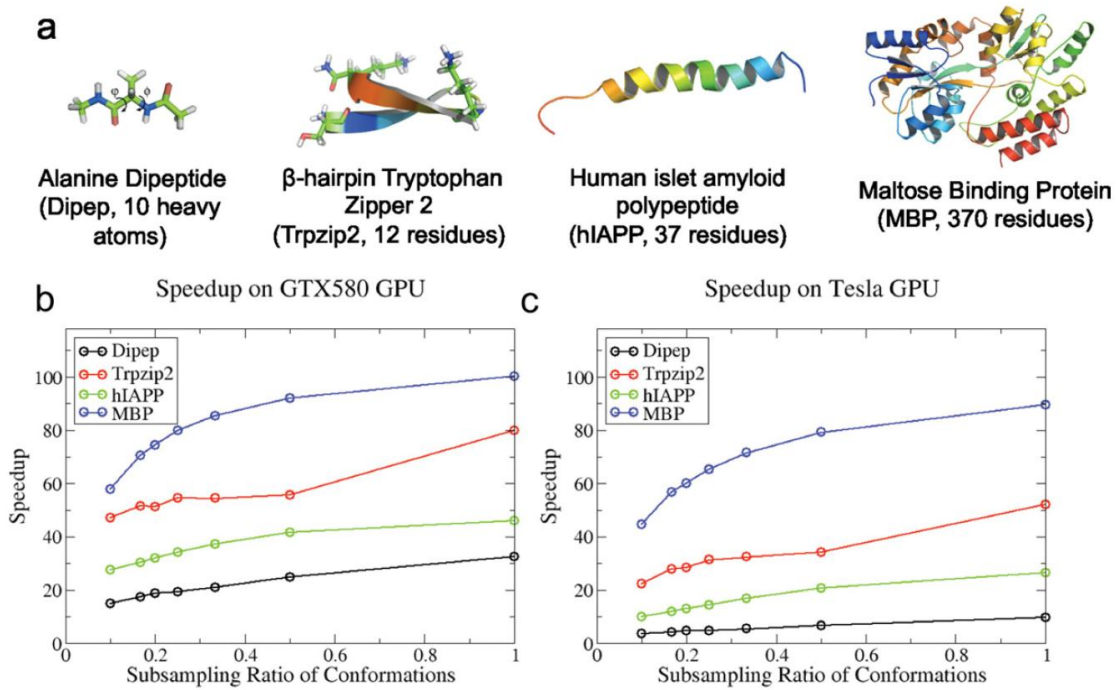


K-Centers Clustering

- The running time the K-centers algorithm is *O(kn)*. With no need for iterations to update the centers, K-centers is much more efficient than K-means (*O(knl)*).
- K-centers' results highly depend on the choice of the 1st center. K-centers is often used for large K (e.g., 1000 or 10000).
- K-centers can be further accelerated on GPUs.



GPU-acceleration of K-centers for clustering peptide conformations (3D-coordinates).

K= 4,000

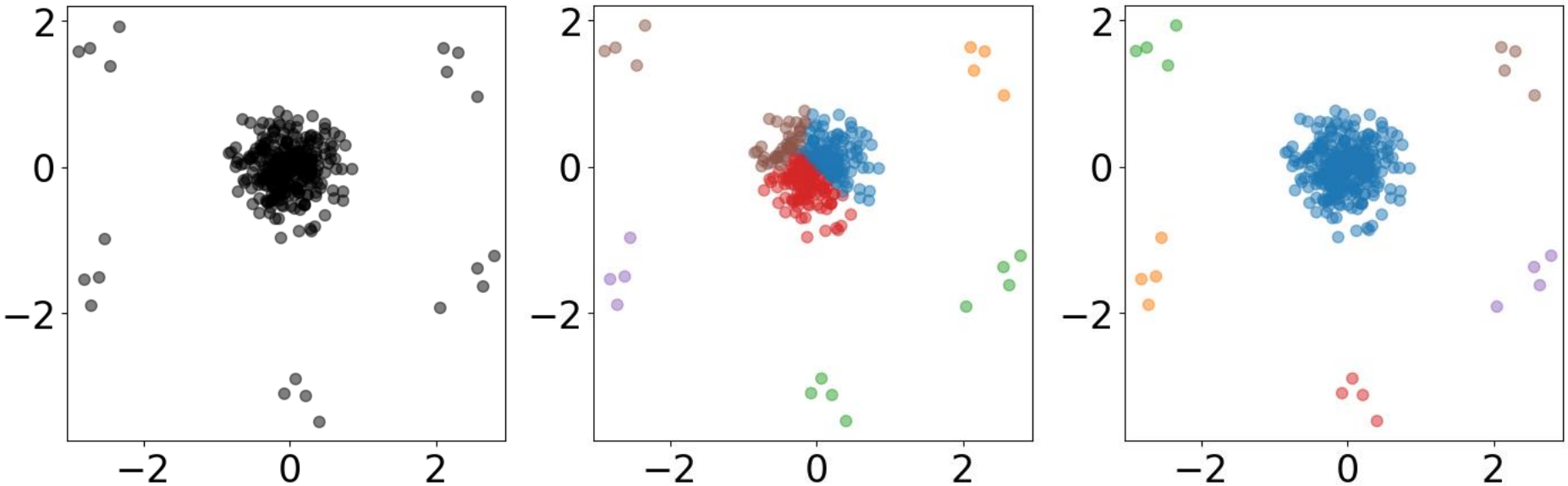
Data size:

- Dipep: n=0.2 million
- Trpzip2: n= 15.6 million
- hIAAP: n=2 million
- MBP: n = 0.25 million

J. Comp. Chem., 34, 95-104, (2013)

Exercise

If we cluster the data points shown below (left panel) into **6** clusters, could you match the results with the applied clustering algorithm (K-means or K-centers)?



(a): ??

(b): ??

Exercise

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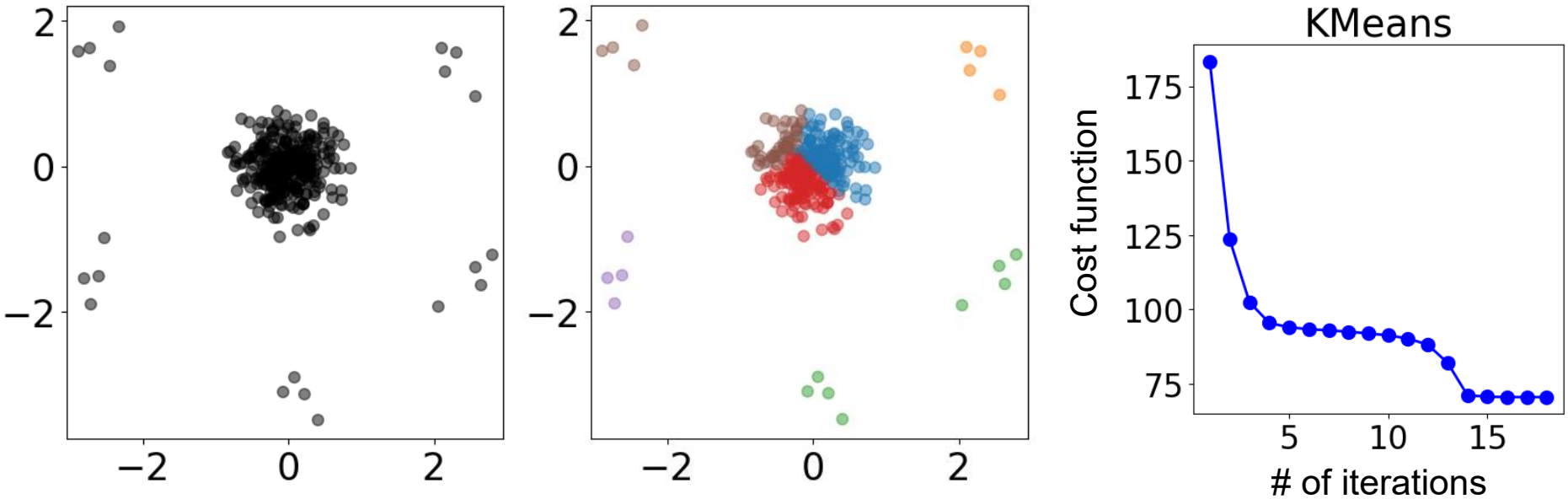
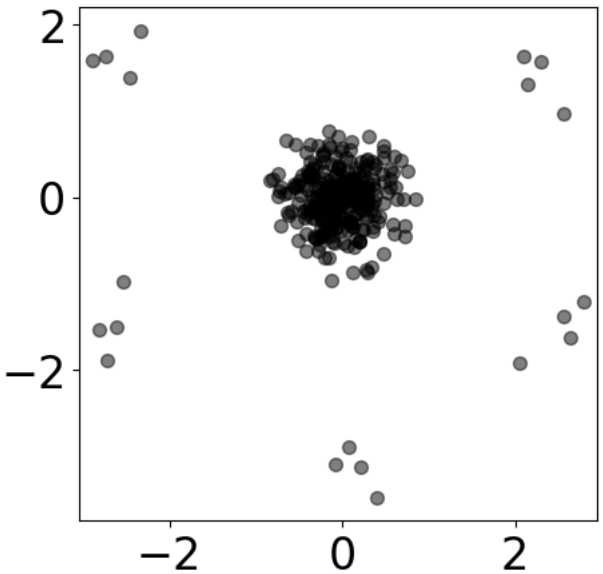


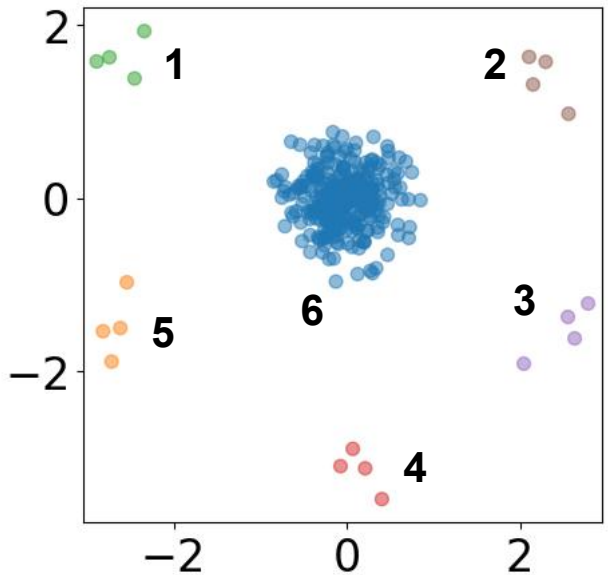
(a): K-means

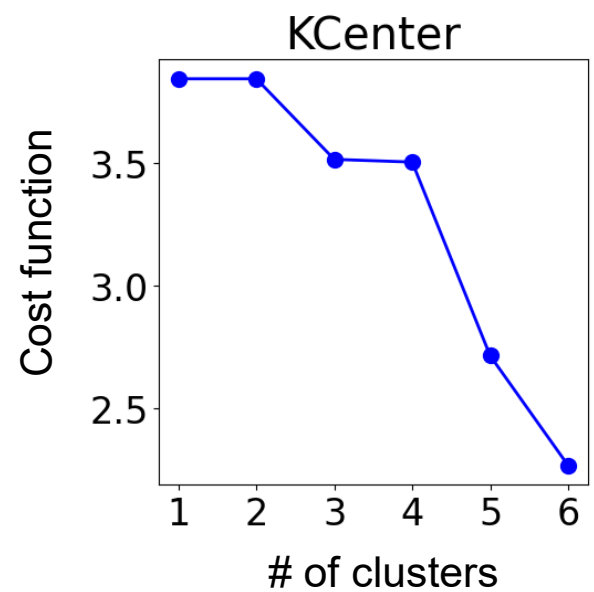
K-means splits the high-density region (free energy minimum).

Exercise

If we cluster the data points shown below (left panel) into 6 clusters, could you match the results with the applied clustering algorithm (K-means or K-centers)?





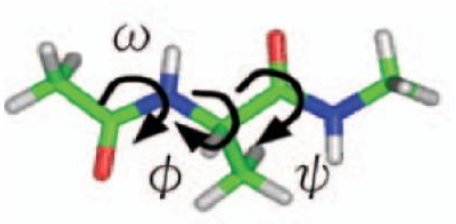


(b): K-centers

K-centers preserves high-density regions within the same cluster (blue), making it preferable for free energy landscape analysis in chemistry.

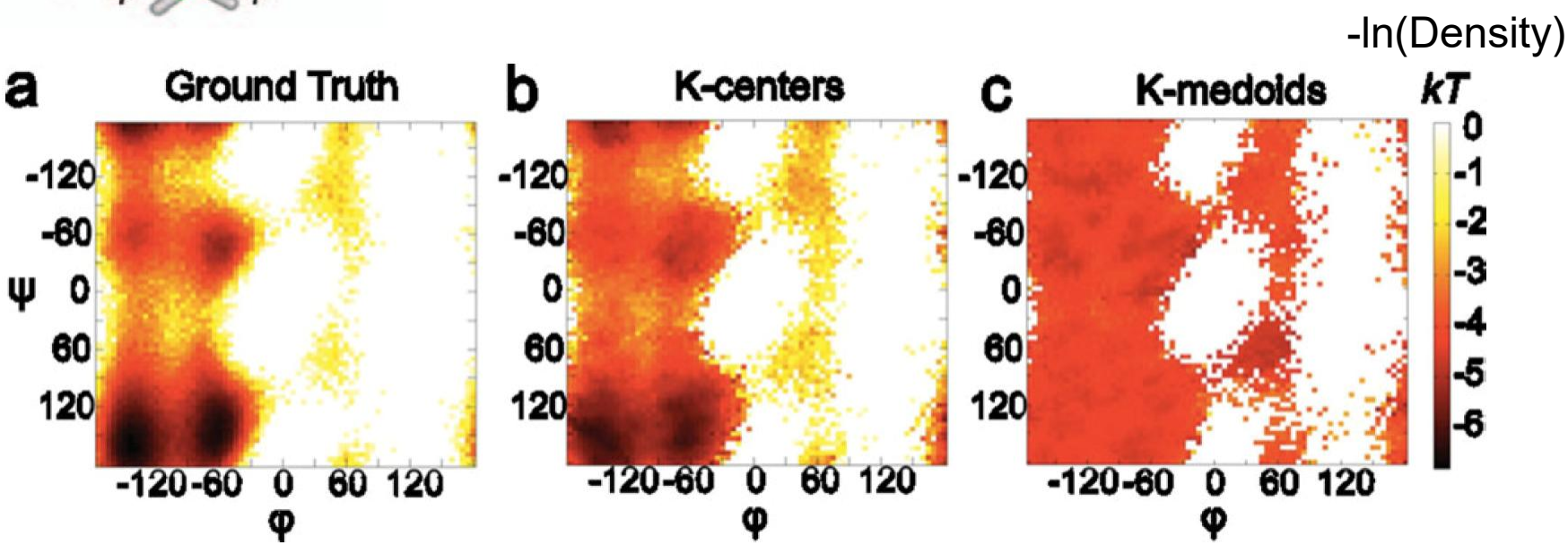
K-centers v.s. K-means

Number of data points in each cluster from K-centers clustering provides a good estimate of the underlying distribution density, a feature absent in K-means (K-medoids).



Cluster 195K conformations into K=4000 clusters.

Density is estimated from fractions of conformations in each cluster.



K-centers in (b) can recover all six minima and their positions as found in the reference density in (a) for alanine dipeptide.

Outline

- Center-based partitional clustering
- **Density-based partitional clustering**
- Hierarchical clustering
- Dimensionality reduction

DBSCAN

Density-Based Spatial Clustering of Applications with Noise (DBSCAN) is a clustering algorithm that groups data points in high-density regions into clusters, while treating data points in low-density regions as outliers or noise.

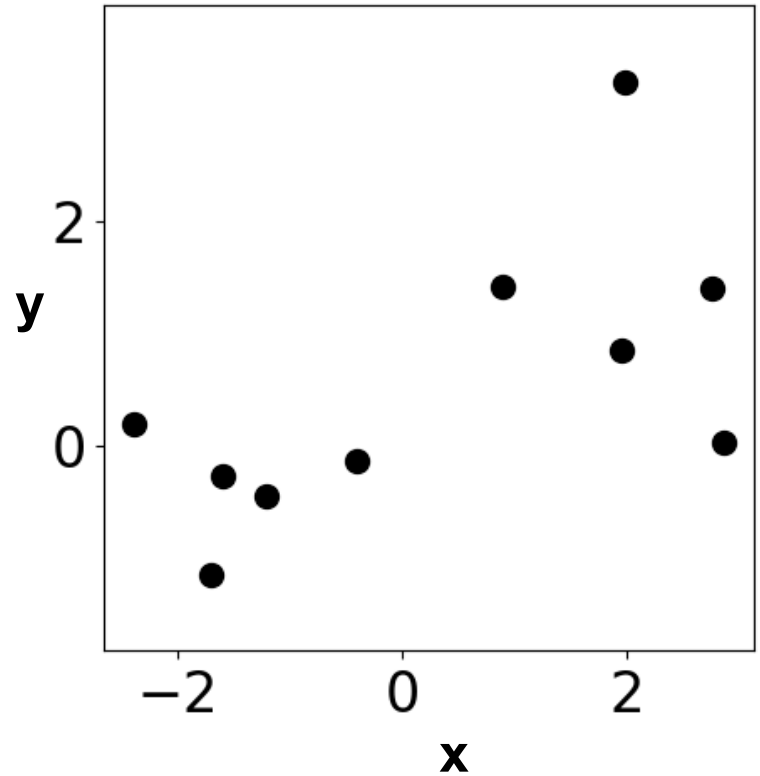
DBSCAN has two hyperparameters:

- Neighboring distance (ε)
- Minimum number of neighboring points (minPts)

Partitional clustering algorithms (K-means, K-centers) assign all data points to clusters, while DBSCAN leave some data points as outliers (not assigned to any cluster).

KDD-96, Portland, OR, (1996)

Task: Cluster 10 data points using DBSCA ($\varepsilon = 1.0$ and $minPts = 2$).



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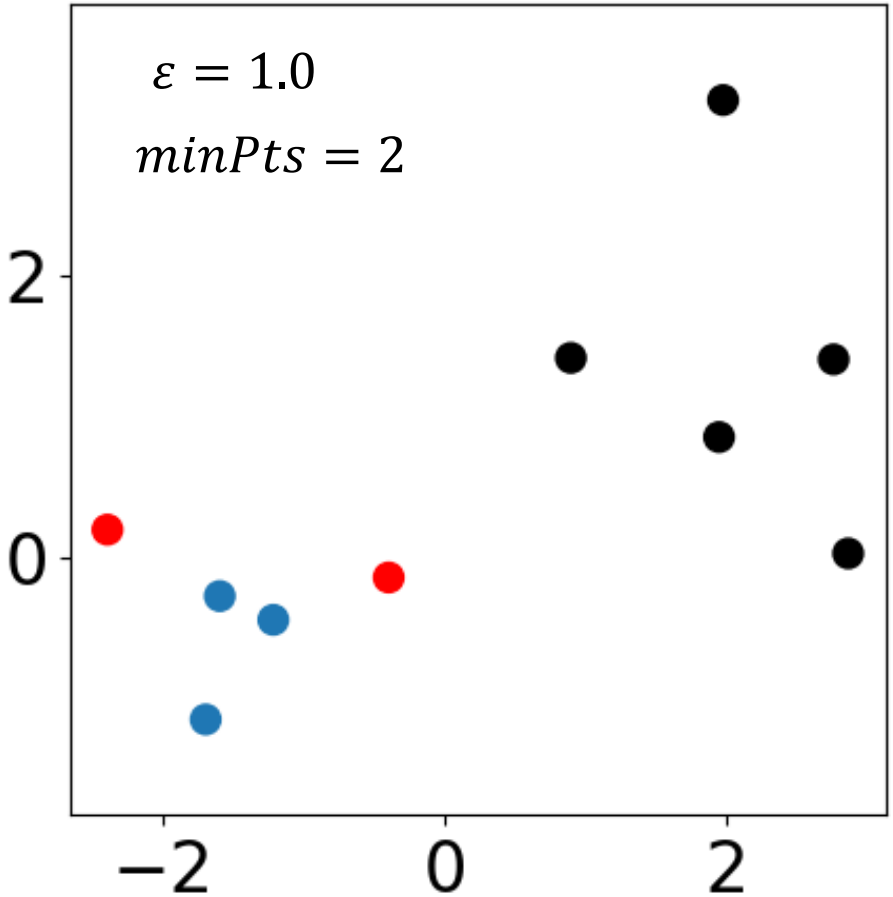
DBSCAN

Step 1: Identify core points, boarder points and noise points.



- **Core points** have at least *minPts*=2 neighbors within the ε-neighborhood.
- **Border points** are within the ε-neighborhood of a core point but have fewer than *minPts* neighboring.
- The remaining points are **noise points**.

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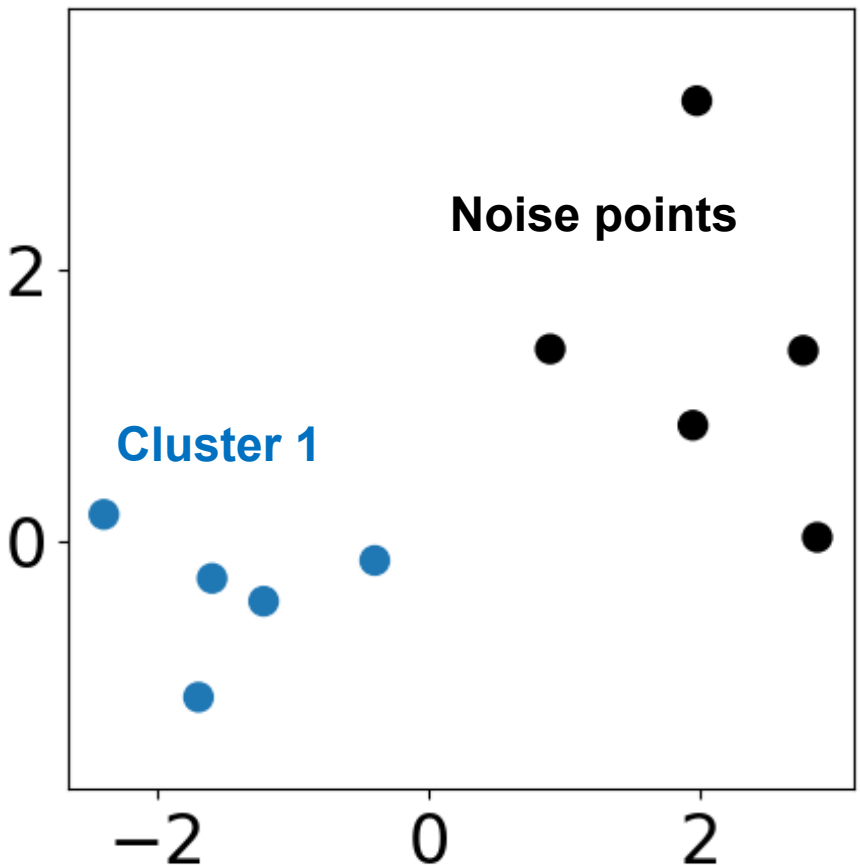
DBSCAN

Step 2: Merge connected core points into a single cluster, and assign border points to the nearest core point.



Two core points are grouped into a single cluster if their distance is less than ε or if they share a border point.

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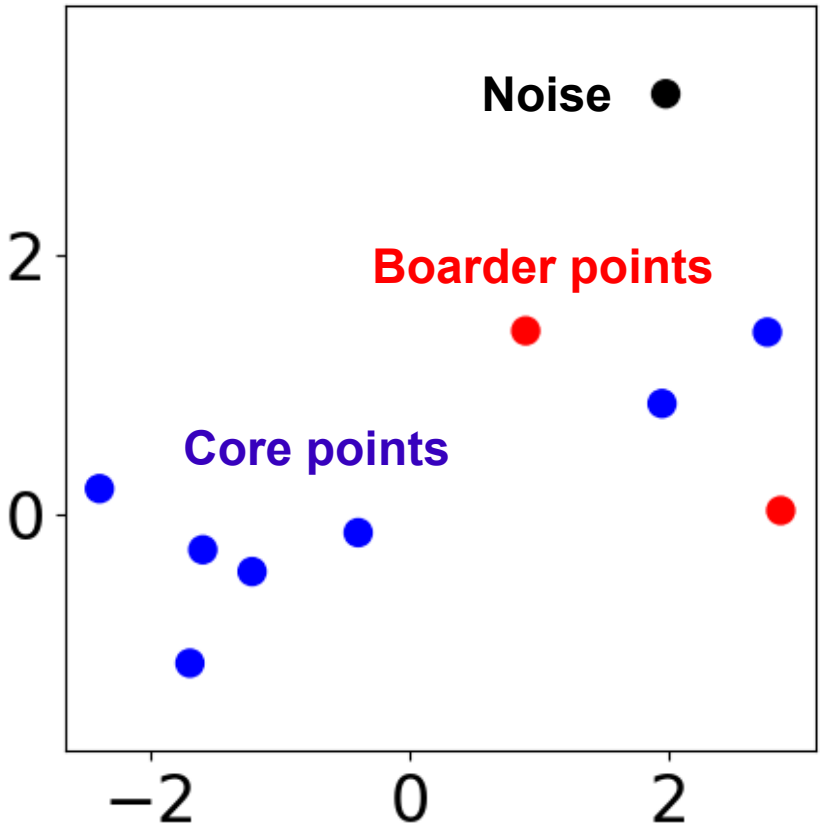
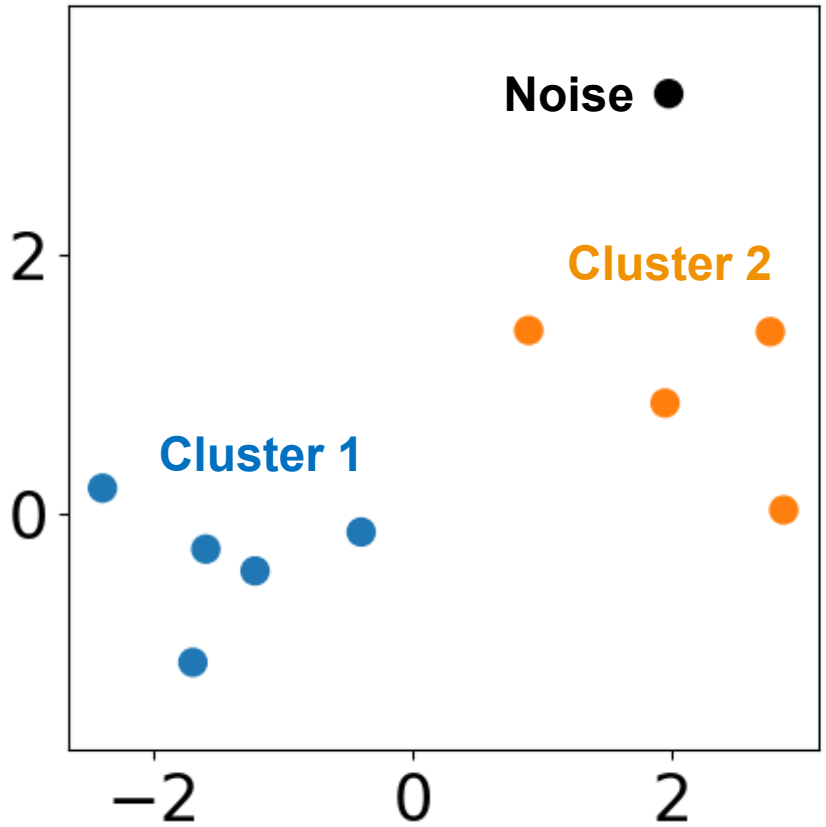
DBSCAN

A new set of hyperparameters: $\varepsilon = 2.0\,,\ \ minPts = 3$





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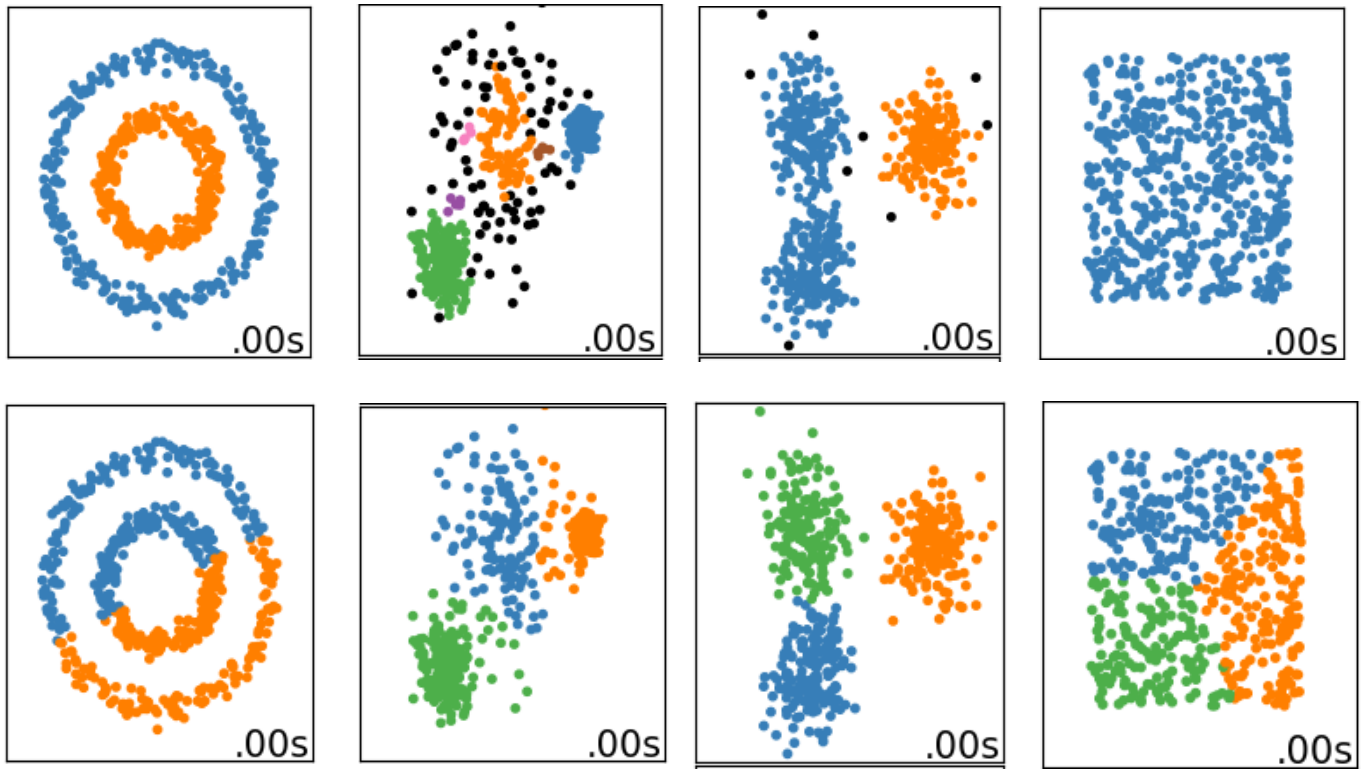
DBSCAN

- DBSCAN clusters points in the same high-density region together, effectively identifying free energy basins.
- Requires no input for the number of clusters.
- Outliers reveal low-density regions (e.g., transition state).

DBSCAN

K-means



https://scikit-learn.org/

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Exercise

After clustering a chemical compound database, our goal is to select **representative chemical compounds** from high-density regions. Which of the following strategies would be appropriate?

(a) Select cluster centers identified by K-medoids clustering.

(b) Select cluster centers identified by K-means clustering.

(c) Select cluster centers identified by K-centers clustering.

(d) Select core points identified by DBSCAN

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Exercise

After clustering a chemical compound database, our goal is to select representative chemical compounds from high-density regions. Which of the following strategies would be appropriate?

(a) Select cluster centers identified by K-medoids clustering.

(b) Select cluster centers identified by K-means clustering.

- Cluster centers may not correspond to real compounds

(c) Select cluster centers identified by K-centers clustering.

- Cluster centers are not located at the center of the cluster.

(d) Select core points identified by DBSCAN

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Exercise

Which clustering algorithms are appropriate for identifying transition state conformations from a conformational ensemble sampled from the Boltzmann distributions?

(a) K-medoids

(b) K-means

(c) K-centers

(d) DBSCAN

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Exercise

Which clustering algorithms are appropriate for identifying transition state conformations from a conformational ensemble sampled from the Boltzmann distributions?

(a) K-medoids

(b) K-means

(c) K-centers

(d) DBSCAN

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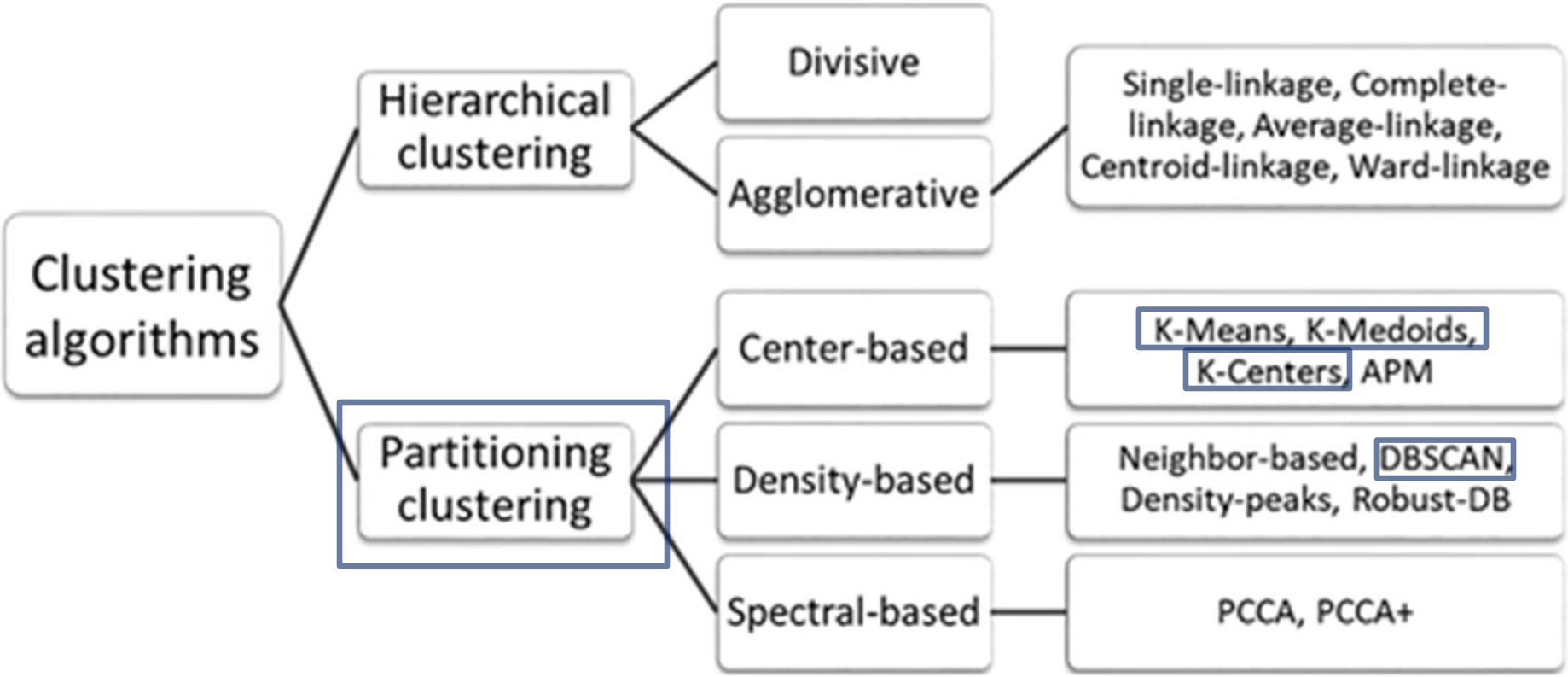
Review of the Previous Lecture

Common clustering algorithms for analyzing chemical data:



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Outline

- Center-based partitional clustering
- Density-based partitional clustering
- **Hierarchical clustering**
- Dimensionality reduction

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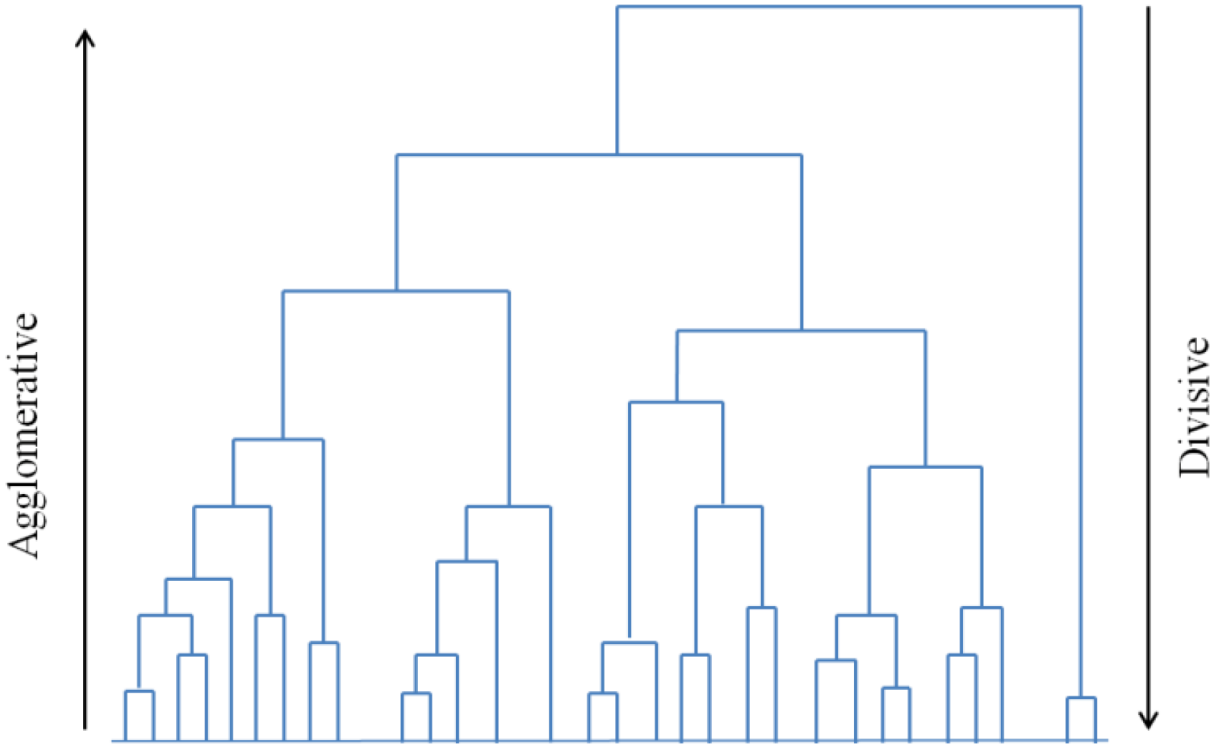
Hierarchical Clustering

- Arrange the input data points in a hierarchy (a binary tree).
- No need to input number of clusters.
- Two types: Bottom-up v.s. Top-down.

Agglomerative (bottom-up): Start with each point as a cluster and progressively merge them.



Divisive (top-down): Start with all points in one cluster and progressively split them.

Neurocomputing 267, 664 (2017).

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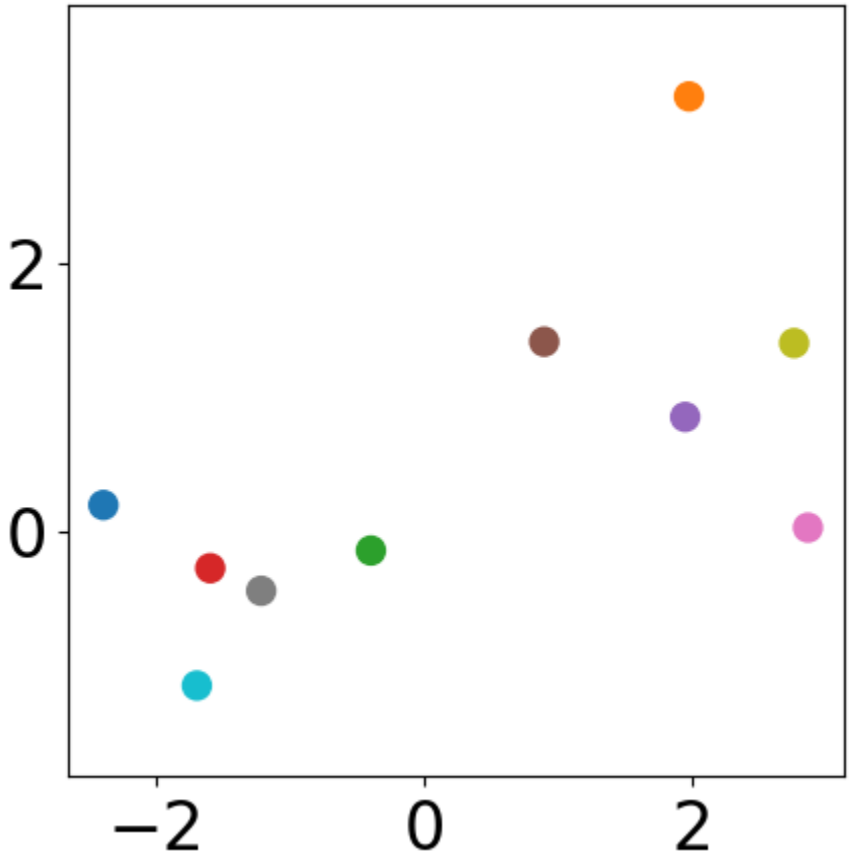
Agglomerative Clustering: Single Linkage

Bottom-up: Merge the two closest clusters at each step into a larger cluster.

Step 1: Each data point starts in its own cluster



10 data points –
10 clusters

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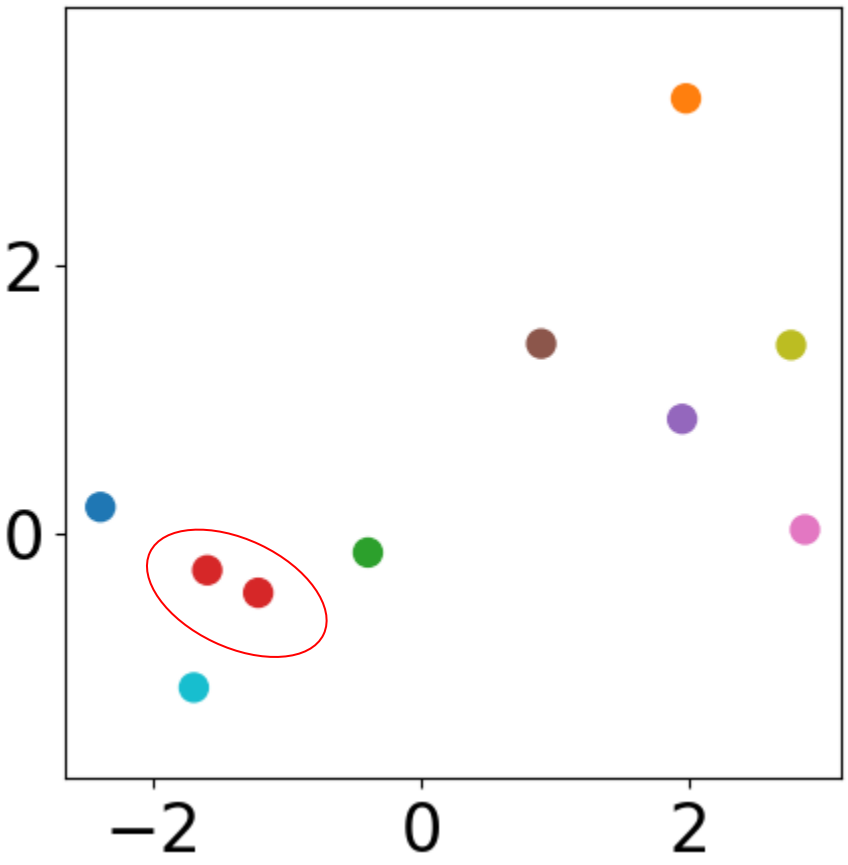
Agglomerative Clustering: Single Linkage

Step 2: Compute distances between clusters and merge the two closest clusters into a larger one.



# of clusters:

10 reduced to 9

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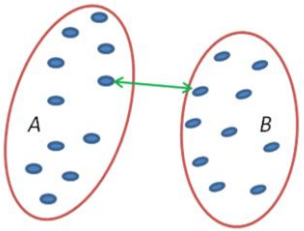
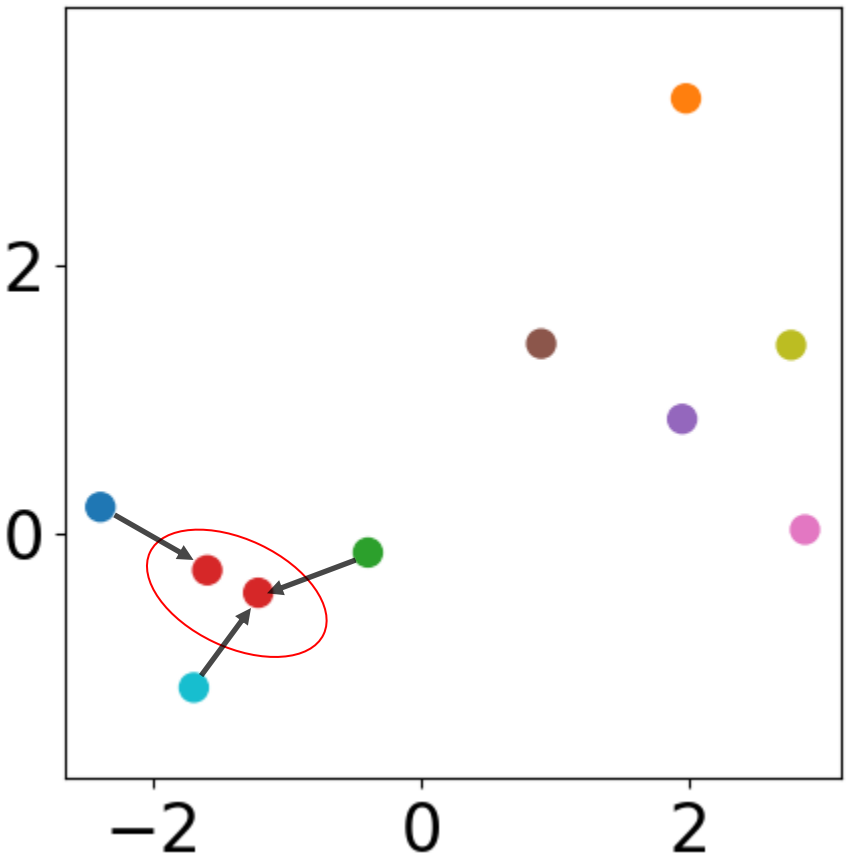
Agglomerative Clustering: Single Linkage

Step 2: Compute distances between clusters and merge the two closest clusters into a larger one.

<u>Distance between 2 clusters</u>: the shortest distance between any pair of points, 1 from each cluster.





# of clusters:

10 reduced to 9

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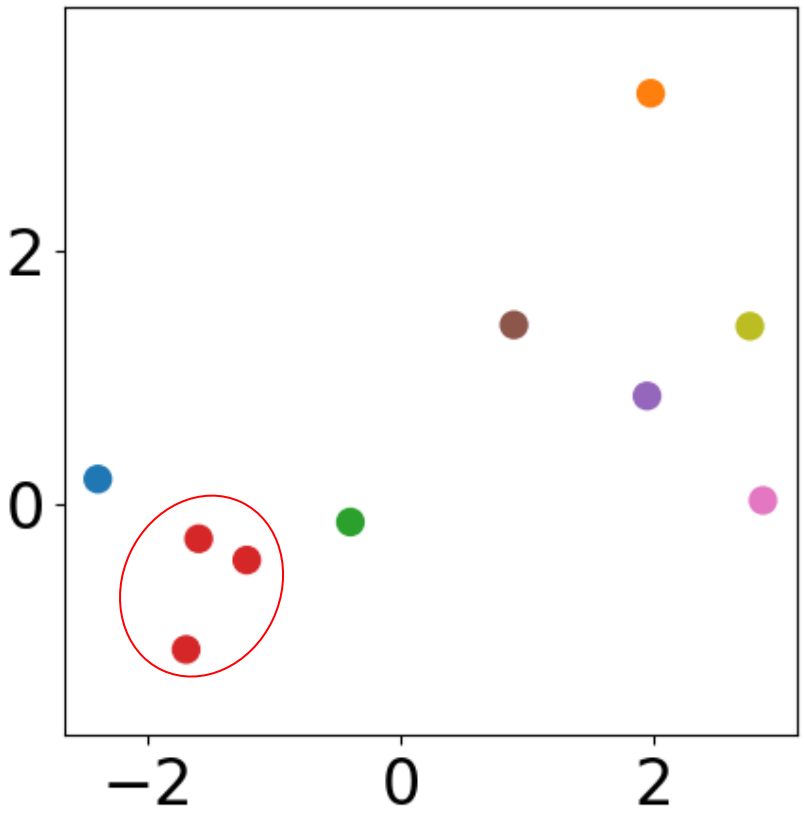
Agglomerative Clustering: Single Linkage

Step 3: 9 clusters merged into 8 clusters.



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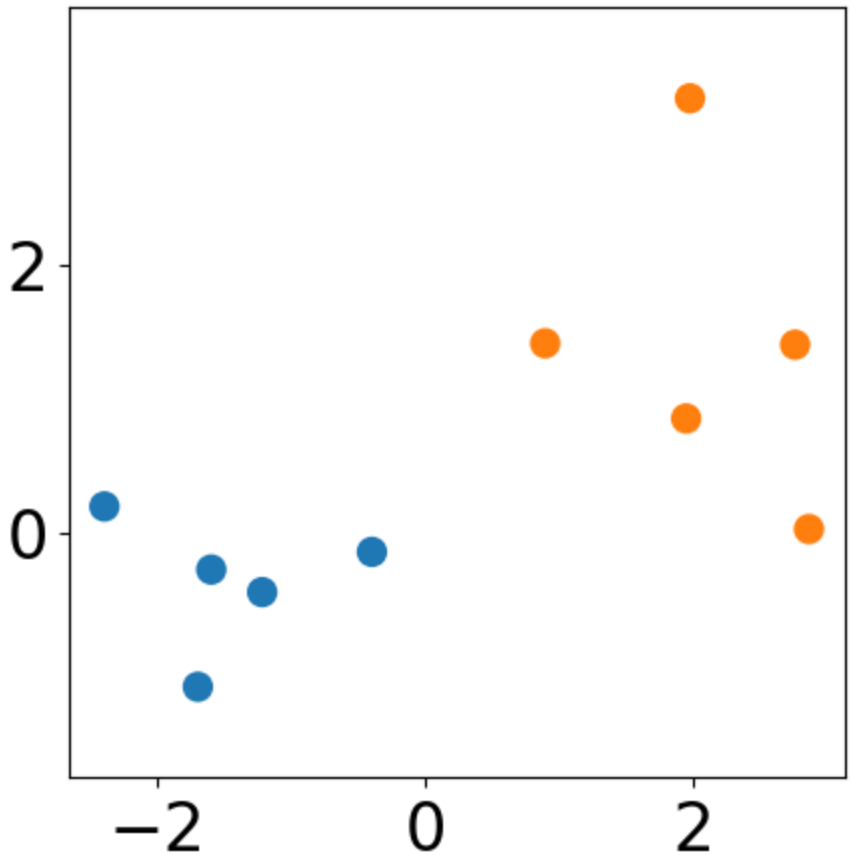
Agglomerative Clustering: Single Linkage

Repeat meagering until the number of centers reaches the hyperparameter K = 2.



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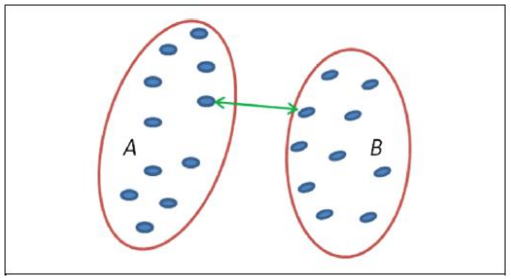
Agglomerative Clustering

Linkage clustering algorithms vary in how they define the distance between pairs of clusters (d_{12}):

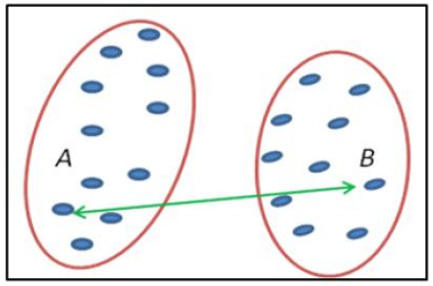
TABLE I Summary of agglomerative hierarchical clustering algorithms.

Subtype	Distance between two clusters[a,b]	Linkage criterion
Single-linkage[a]	$d_{12}=\min_{i,j}(d(C1_i, C2_j))$	minimum d_{12}
Complete-linkage[a]	$d_{12}=\max_{i,j}(d(C1_i, C2_j))$	minimum d_{12}
Average-linkage[a]	$d_{12}=\mathrm{mean}_{i,j}(d(C1_i, C2_j))$	minimum d_{12}
Centroid-linkage[b]	$d_{12}=d(\overline{C1}, \overline{C2})$	minimum d_{12}
Ward-linkage[c]	$d_{12}=\sum_i d(\{C1 \cup C2\}_i, \overline{\{C1 \cup C2\}})$	minimum increase in d_{12}

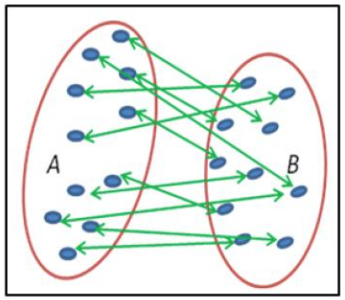
[a] $d(C1_i, C2_j)$ denotes to the distance between the ith element in cluster $C1$ and the jth element in cluster $C2$.
[b] $\overline{C}$ represents the geometrical mean of a certain cluster C.
[c] d_{12} denotes to the total within-cluster variance of the union of two clusters $C1 \cup C2$.



Single-linkage



Complete-linkage



Average-linkage

Neurocomputing 267, 664 (2017) *Chin. J. Chem. Phys*. 31, 404 (2018)

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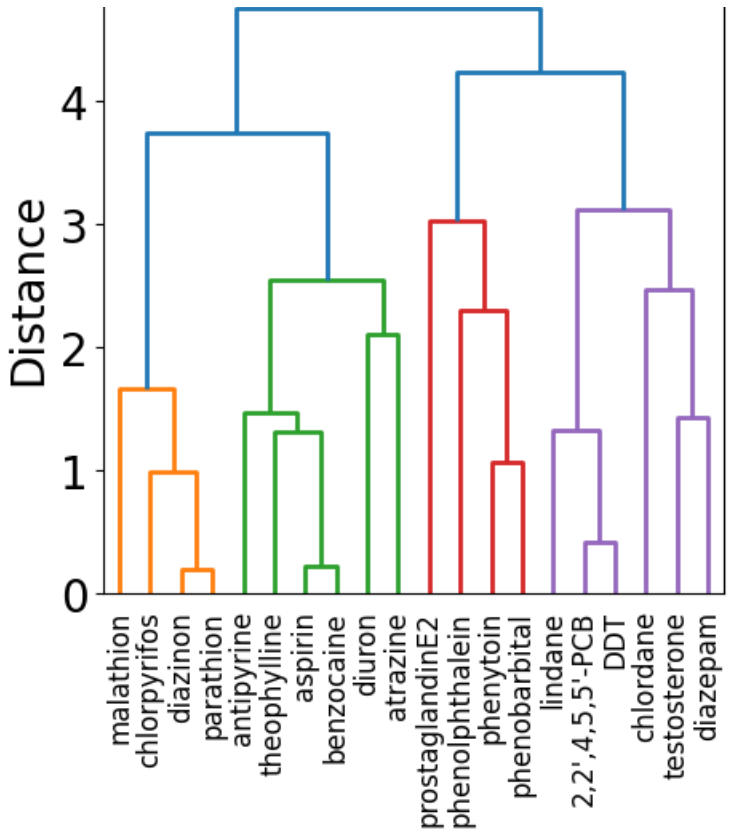
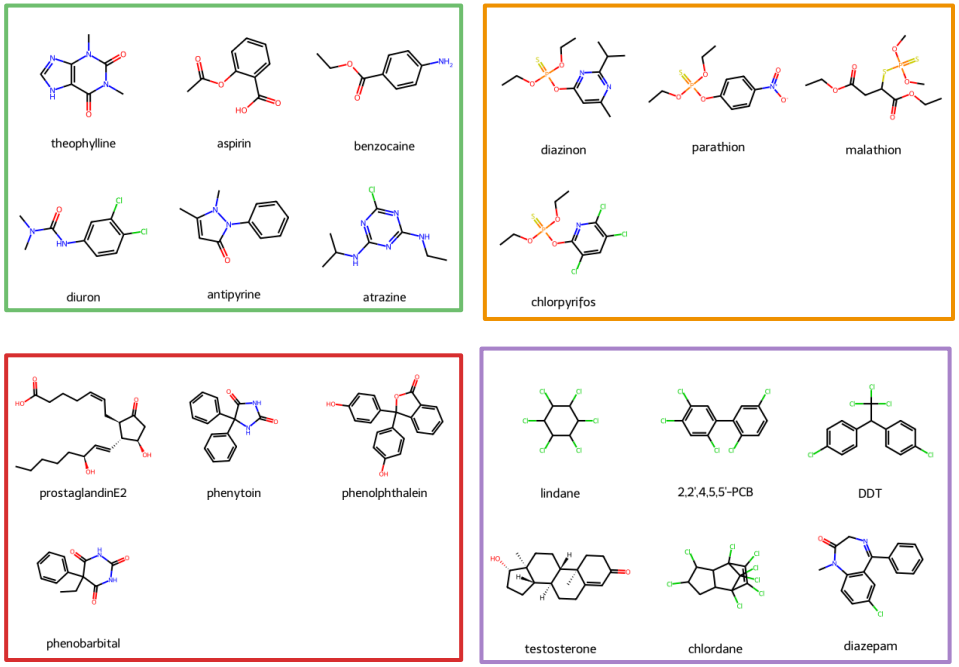
Complete-linkage Clustering on Chemical Compounds

Euclidean distance between feature vectors:

[MW, Hdonor, HAcceptor, Rings]





Four clusters

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Exercise

What is the maximum depth of the binary tree resulting from hierarchical clustering on n data points?

(a). 2

(b). n

(c). n/2

(d). n-1

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Exercise

What is the maximum depth of the binary tree resulting from hierarchical clustering on n data points?

(a). 2

(b). n

(c). n/2

(d). n-1

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Outline

- Center-based partitional clustering
- Density-based partitional clustering
- Hierarchical clustering
- **Dimensionality reduction**

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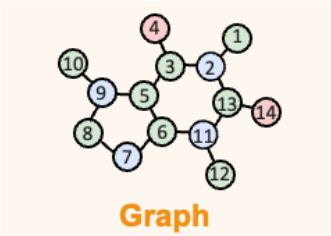
High-Dimensional Chemical Data

- Chemical data is high-dimensional with lots of features.
- **Example:** The PubChem database contains >119.3 million chemical compounds, each with numerous molecular properties.



Generic names	Caffeine, Guaranine, 1,3,7-Trimethylxanthine
Molecular Weight	194.19
# of H-Bond Donors	0
# of H-Bond Acceptors	6 **Molecular Properties**
# of Rotable Bonds	0
logP (octanol-water partition coefficient)	-1.03
Molecular Formula	$C_8H_{10}N_4O_2$ **Strings**
SMILES	Cn1c(=O)c2c(ncn2C)n(C)c1=O
InChI (International Chemical Identifier)	InChI=1S/C8H10N4O2/c1-10-4-9-6-5(10)7(13)12(3)8(14)11(6)2/h4H,1-3H3

- **Example:** RCSB PDB PROTEIN DATA BANK **230,083 structures by Jan 2025**

A protein containing 300 residues have ~30K atoms ✕ 3 = 90K coordinates.

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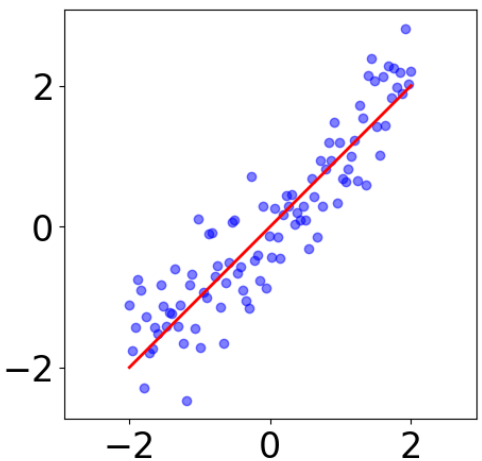
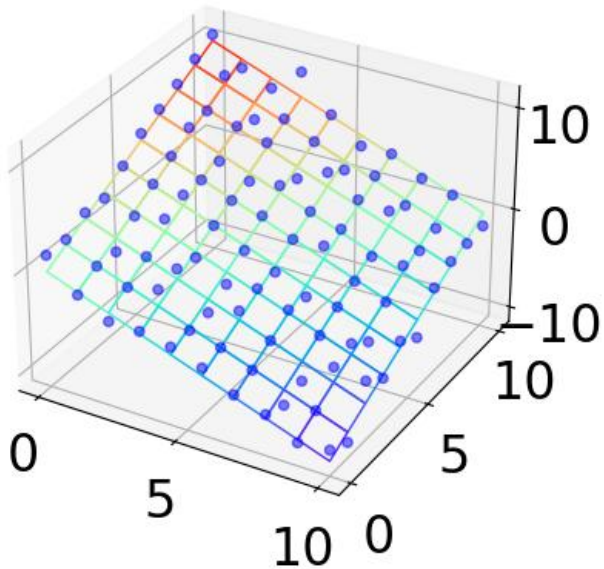
Discover Low Dimensional Structure

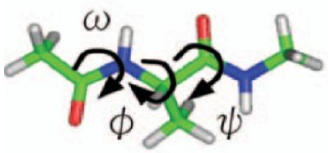
Principal Component Analysis (PCA) reduces dimensionality and identifies the low-dimensional linear subspace of the data.



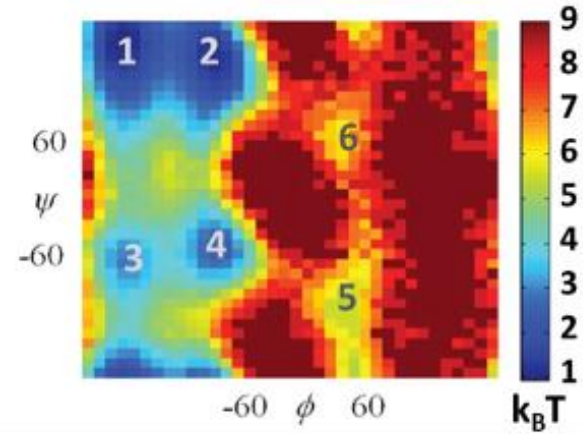
$D = 2$
$d = 1$



$D = 3$
$d = 2$



22 atoms
66 coordinates



$D = 66$
$d = 2$

High-dimensional data often resides on lower-dimensional spaces or manifolds.

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PCA

100 data points in a 2-dimenstional space:

$$\begin{array}{c} \\ 2 \times \\ 100 \\ \text{matrix} \end{array} \begin{array}{c} \begin{array}{cccc} x_1 & x_2 & & x_{100} \end{array} \\ \begin{bmatrix} -2.00 & -1.96 & \dots & 2.00 \\ -1.12 & -1.76 & \dots & 2.20 \end{bmatrix} \end{array}$$

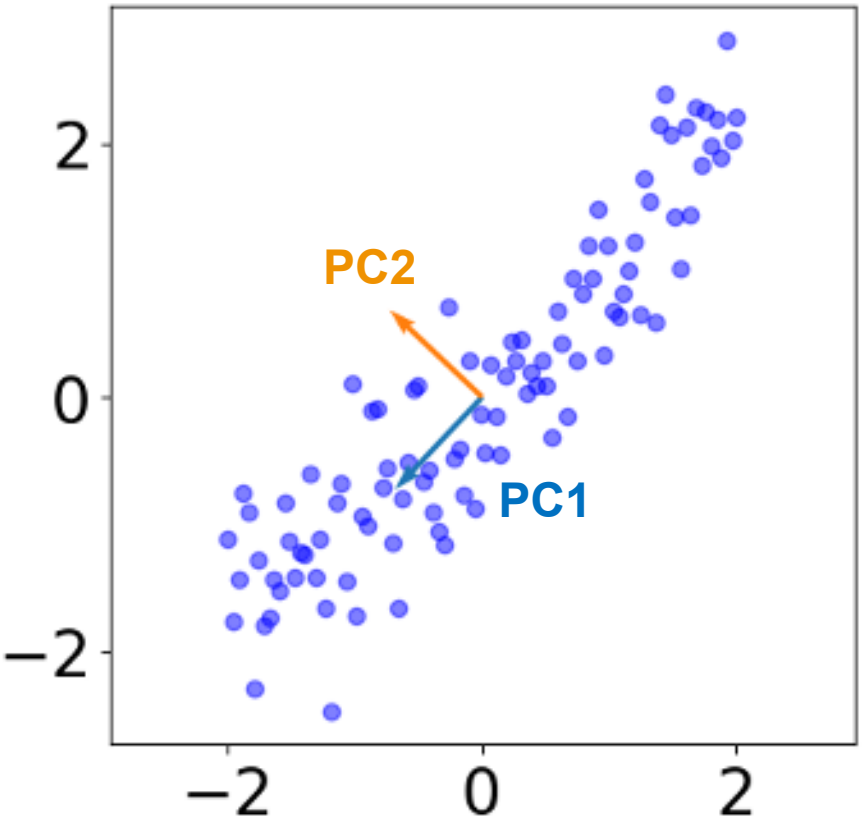
$$mean = \begin{bmatrix} 0 \\ 0.03 \end{bmatrix}$$

Remove mean

$$X = \begin{bmatrix} -2.00 & -1.96 & \dots & 2.00 \\ -1.15 & -1.79 & \dots & 2.17 \end{bmatrix}$$

Compute **covariance matrix** (XX^T)

$$XX^T = \begin{bmatrix} 136.03 & 130.97 \\ 130.97 & 151.31 \end{bmatrix}$$ 2×2 matrix



Eigen-decomposition of the covariance matrix:

$$XX^T v = \lambda v$$

$\lambda_1 = 274.86 \quad v_1 = \begin{bmatrix} -0.69 \\ -0.73 \end{bmatrix}$: Principal component 1 (**PC1**)

$\lambda_2 = 12.48 \quad v_2 = \begin{bmatrix} -0.73 \\ 0.68 \end{bmatrix}$: Principal component 2 (**PC2**)

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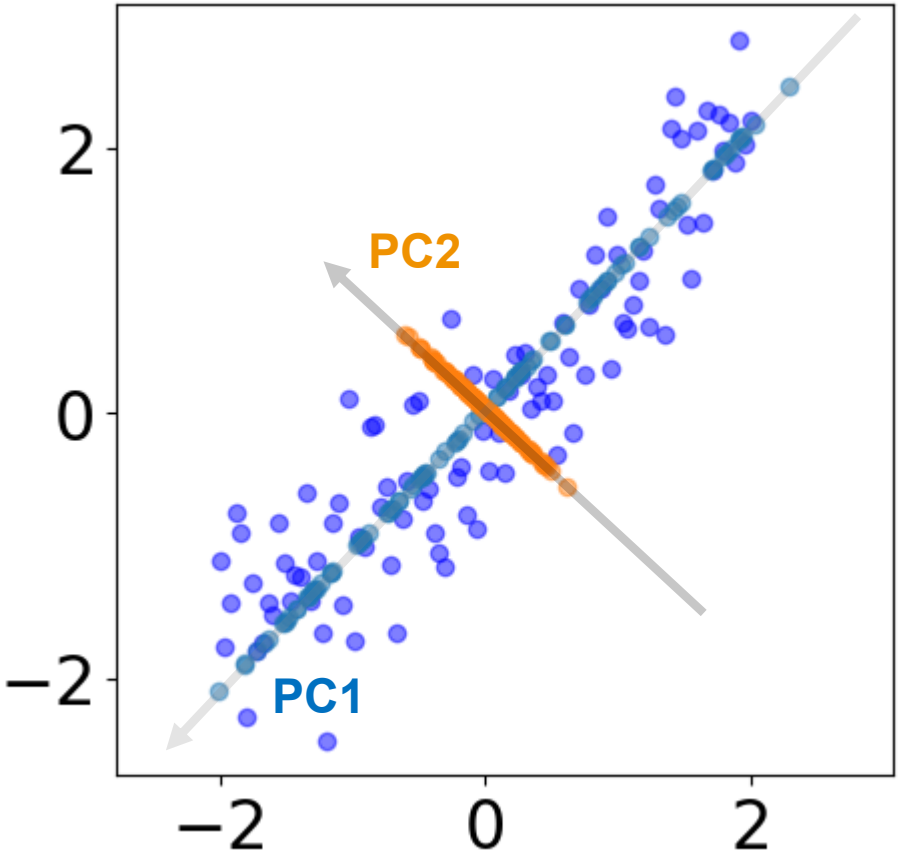
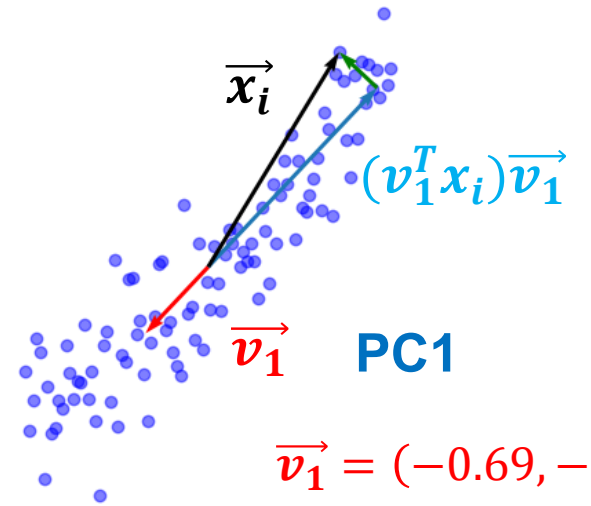
PCA

2-dimensional data is reduced to 1 dimension:



- **Projection data points onto PC1**
- **Projection data points onto PC2**



$\vec{v_1} = (-0.69, -0.73)$

$$x_i = \begin{bmatrix} 1.43 \\ 2.35 \end{bmatrix} \qquad v_1 = \begin{bmatrix} -0.69 \\ -0.73 \end{bmatrix}$$

$$v_1{}^T x_i = -2.69 \qquad (v_1{}^T x_i)\vec{v_1} = (1.85, 1.96)$$

How about projecting x_i on PC2?

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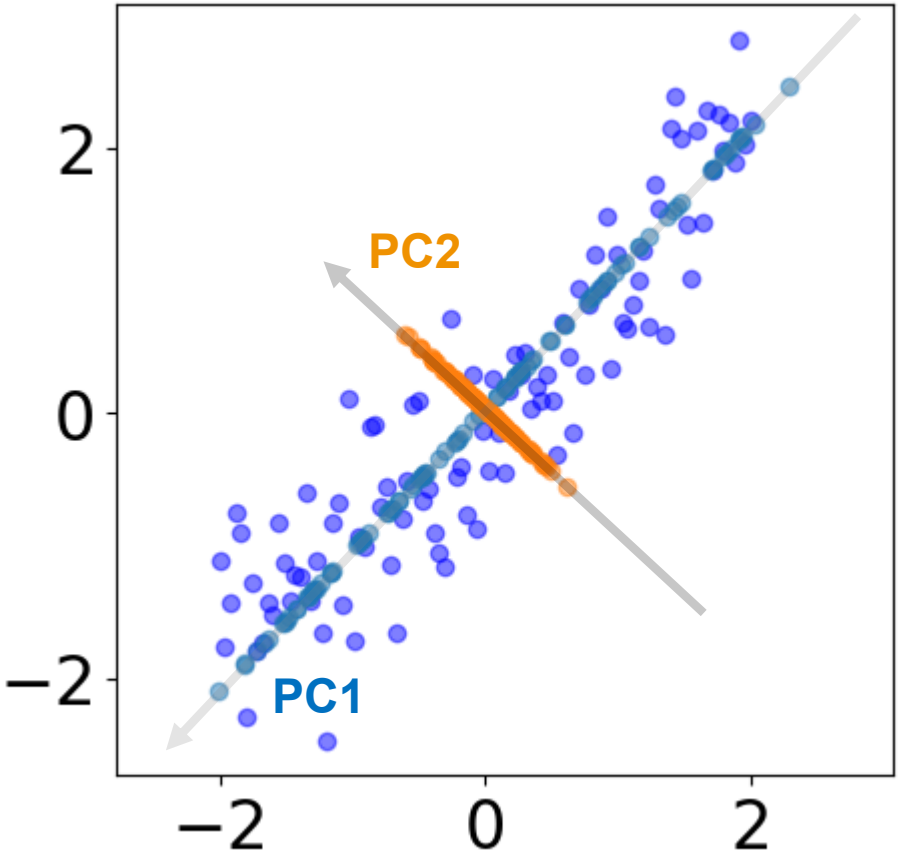
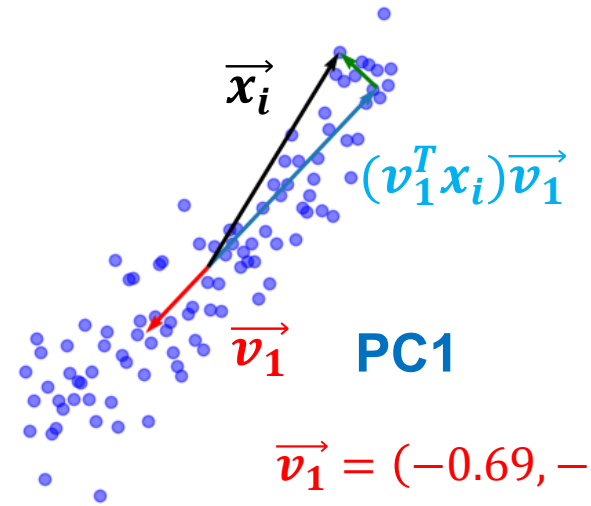
PCA

2-dimensional data is reduced to 1 dimension:



- **Projection data points onto PC1**
- **Projection data points onto PC2**



$\vec{v_1} = (-0.69, -0.73)$

$x_i = \begin{bmatrix} 1.43 \\ 2.35 \end{bmatrix}$ $\quad v_1 = \begin{bmatrix} -0.69 \\ -0.73 \end{bmatrix}$

${v_1}^T x_i = -2.69$ $\quad (v_1{}^T x_i)\vec{v_1} = (1.85, 1.96)$

How about projecting x_i on PC2? ${v_2}^T x_i = 0.5541$

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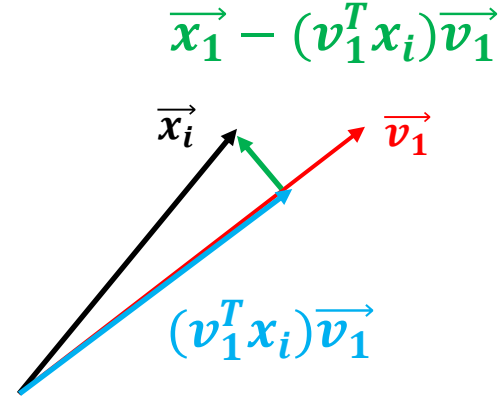
PCA Theory

The first principal component (PC1 = **v**)

- **Minimizes the reconstruction error:**

Reconstruction error:

$$\sum_{i=1}^{n} \|\mathbf{x}_i - (\mathbf{v}^T\mathbf{x}_i)\mathbf{v}\|^2$$



- Follows the direction that **maximizes the variance:**

Variance:

$$\sum_{i=1}^{n} (\mathbf{v}^T\mathbf{x}_i)^2 = \mathbf{v}^T\mathbf{X}\mathbf{X}^T\mathbf{v}$$

- Maximize variance: Maximize the **blue** line
- Minimize the reconstruction error: minimize the **green** line.

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Variance and Covariance

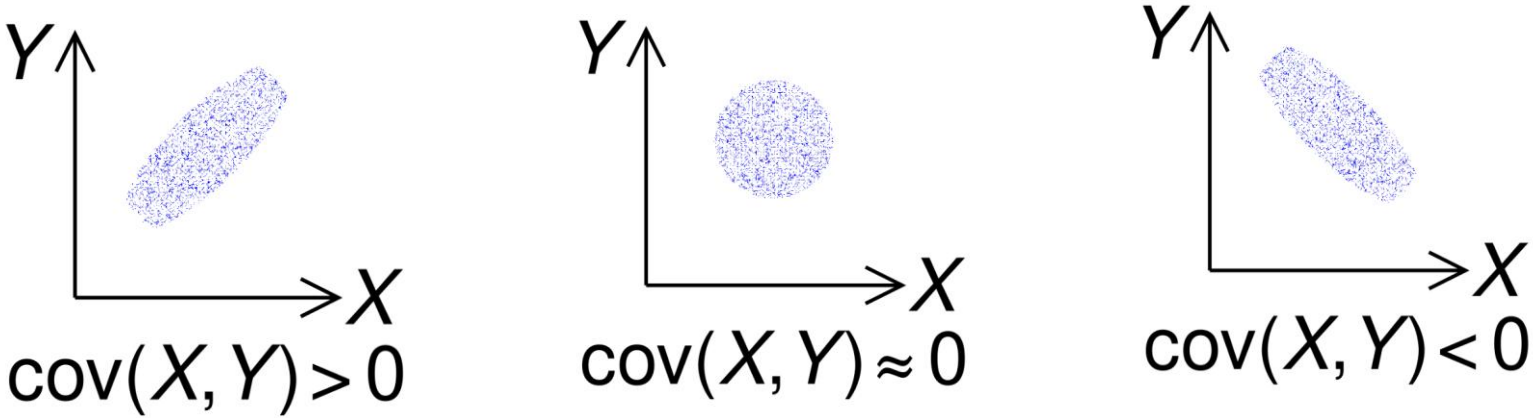
Principal components (PCs) are eigenvectors of the **covariance** matrix:

- **Variance** measures the spread between values of a feature in a data set.

$$\mathrm{V}ar(x) = \frac{1}{n}\sum_{i}^{n}(x_i - \bar{x})^2$$

- **Covariance** measures joint variability of values of two features in a data set.

$$Cov(x, y) = \frac{1}{n}\sum_{i}^{n}(x_i - \bar{x})(y_i - \bar{y})$$



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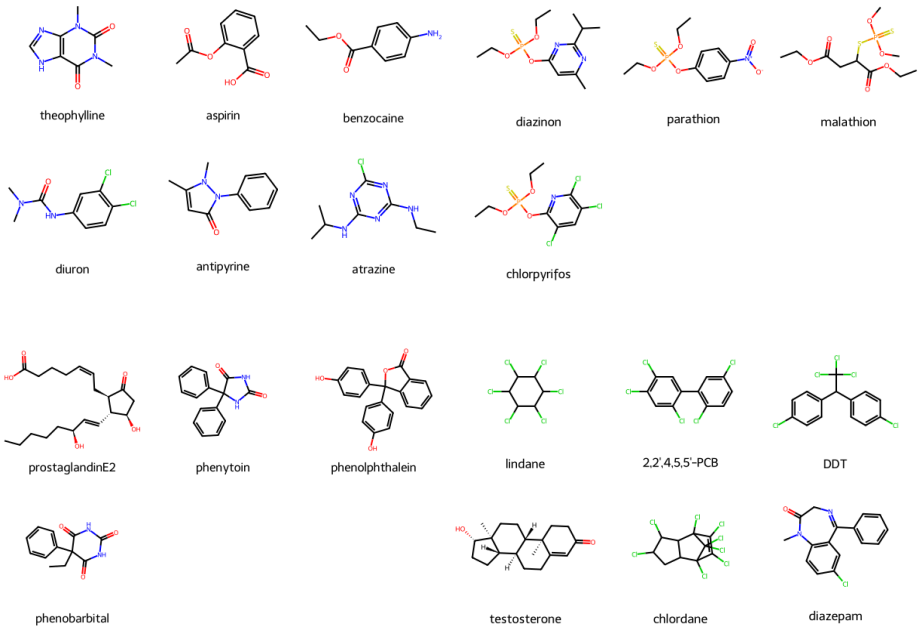
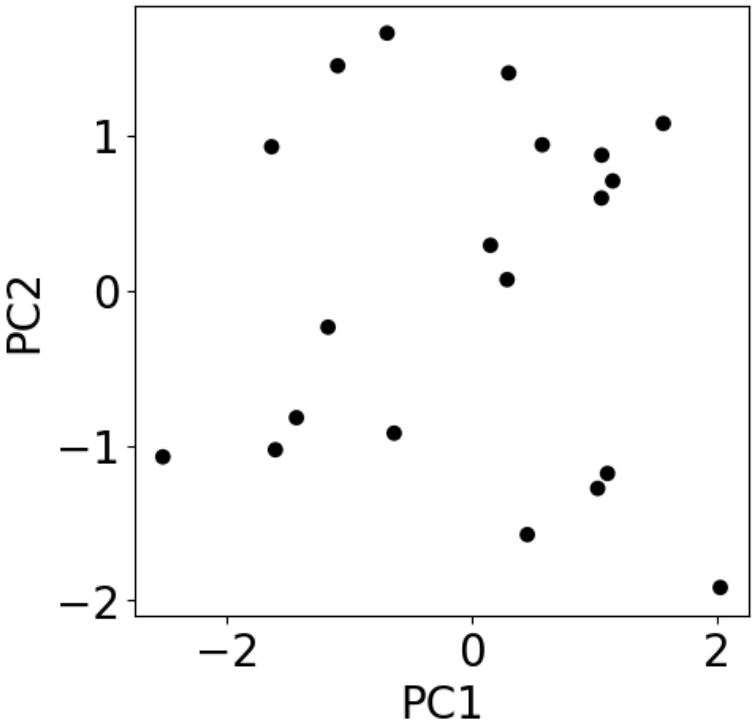
PCA on Chemical Compounds

20 chemical compounds with the feature vector:

[MW, Hdonor, HAcceptor, Rings]



Projections onto 2 PCs:



Reduce the dimension from 4 to 2.

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PCA on Chemical Compounds

$$X = \begin{array}{c} \begin{array}{cccc} \text{Mol 1} & \text{Mol 2} & \dots & \text{Mol 20} \end{array} \\ \begin{bmatrix} 304.35 & 291.27 & \dots & 232.24 \\ 0 & 0 & \dots & 2 \\ 6 & 6 & \dots & 3 \\ 1 & 1 & \dots & 2 \end{bmatrix} \end{array} \begin{array}{l} \text{MW} \\ \text{HBA} \\ \text{HBD} \\ \text{Rings} \end{array} \xrightarrow{\text{Remove mean and normalize}} X = \begin{bmatrix} 0.40 & 0.21 & \dots & -0.67 \\ -0.86 & -0.86 & \dots & 1.30 \\ 1.30 & 1.30 & \dots & -0.04 \\ -0.74 & -0.74 & \dots & 0.19 \end{bmatrix}$$

v.s. **Normalizing** input data **- For each feature**

Calculate the empirical mean and standard deviation for a feature:

Mean: $$\mu_\alpha = \frac{1}{n}\sum_{i=1}^{n} x_\alpha^{(i)}$$

Standard deviation: $$\sigma_\alpha = \left(\frac{1}{n}\sum_{i=1}^{n}\left(x_\alpha^{(i)} - \mu_i\right)^2\right)^{\frac{1}{2}}$$

Normalize input data:

$$\tilde{x}_\alpha^{(j)} = \frac{x_\alpha^{(j)} - \mu_\alpha}{\sigma_\alpha}$$

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PCA on Chemical Compounds

$$X = \begin{array}{c} \begin{array}{cccc} \text{Mol 1} & \text{Mol 2} & \dots & \text{Mol 20} \end{array} \\ \begin{bmatrix} 304.35 & 291.27 & \dots & 232.24 \\ 0 & 0 & \dots & 2 \\ 6 & 6 & \dots & 3 \\ 1 & 1 & \dots & 2 \end{bmatrix} \end{array} \begin{array}{l} \text{MW} \\ \text{HBA} \\ \text{HBD} \\ \text{Rings} \end{array} \xrightarrow[\text{and normalize}]{\text{Remove mean}} X = \begin{bmatrix} 0.40 & 0.21 & \dots & -0.67 \\ -0.86 & -0.86 & \dots & 1.30 \\ 1.30 & 1.30 & \dots & -0.04 \\ -0.74 & -0.74 & \dots & 0.19 \end{bmatrix}$$

4×20 matrix

Covariance matrix (4×4 matrix):

$$XX^T = \begin{bmatrix} 20.00 & -4.77 & -2.59 & 3.12 \\ -4.77 & 20.00 & 2.12 & 3.20 \\ -2.59 & 2.12 & 20 & -8.56 \\ 3.12 & 3.20 & -8.56 & 20 \end{bmatrix}$$

PC1: $v_1 = \begin{bmatrix} -0.41 \\ 0.13 \\ 0.65 \\ -0.62 \end{bmatrix}$ $\lambda_1 = 30.30$

PC2: $v_2 = \begin{bmatrix} -0.50 \\ 0.77 \\ -0.10 \\ 0.39 \end{bmatrix}$ $\lambda_2 = 24.44$

Eigen-decomposition:

$$XX^T v = \lambda v$$

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PCA on Chemical Compounds

$$X = \begin{bmatrix} 0.40 & 0.21 & \dots & -0.67 \\ -0.86 & -0.86 & \dots & 1.30 \\ 1.30 & 1.30 & \dots & -0.04 \\ -0.74 & -0.74 & \dots & 0.19 \end{bmatrix}$$

4×20 matrix

$$V = \begin{bmatrix} -0.41 & -0.50 \\ 0.13 & 0.77 \\ 0.65 & -0.10 \\ -0.62 & 0.39 \end{bmatrix}$$

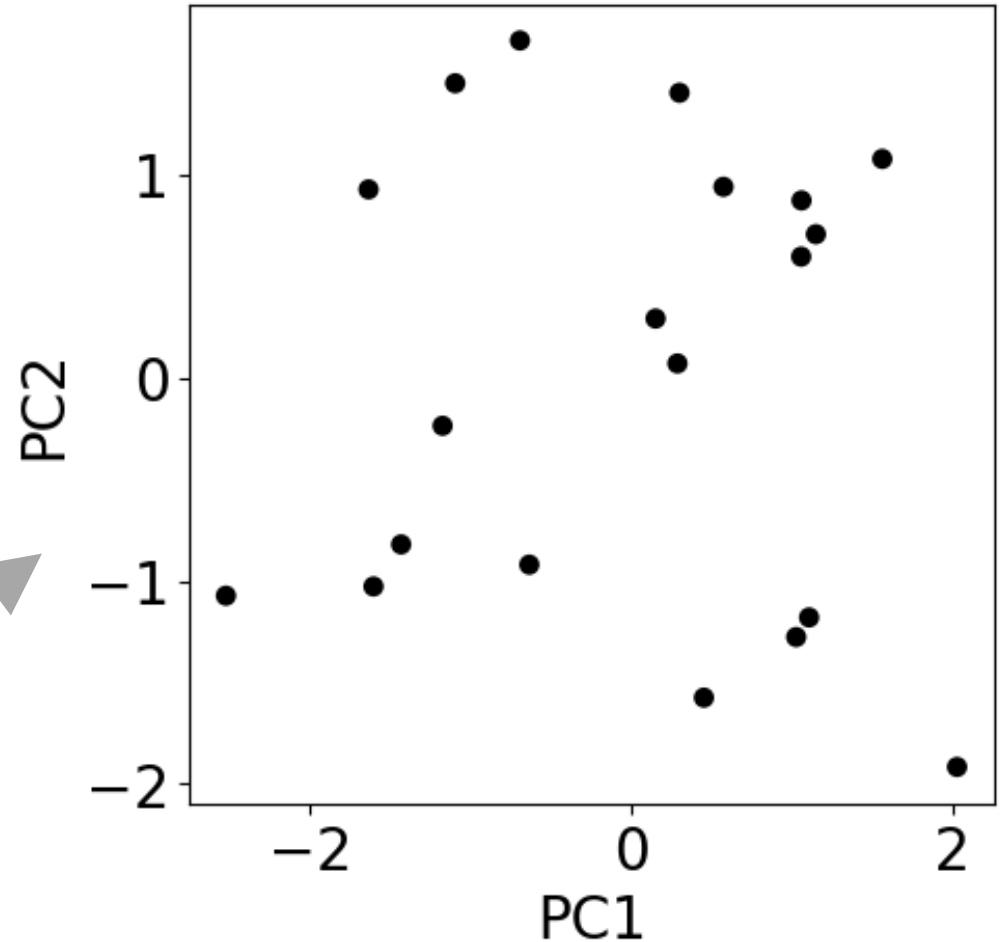
4×2 matrix

Projection onto PCs:

$$\begin{array}{cccc} \text{Mol 1} & \text{Mol 2} & \dots & \text{Mol 20} \end{array}$$

$$V^T X = \begin{bmatrix} 1.02 & 1.11 & \dots & 0.30 \\ -1.28 & -1.18 & \dots & 1.41 \end{bmatrix}$$

2×20 Matrix



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PCA

- The first principal component (PC1) is the leading eigenvector of the covariance matrix XX^T,

- Its eigenvalue λ indicates the amount of variability captured along PC1

$$v_1 = \begin{bmatrix} -0.41 \\ 0.13 \\ 0.65 \\ -0.62 \end{bmatrix} \qquad \lambda_1 = 30.30$$

- PCs are eigenvectors of the covariance matrix. $\boxed{\mathrm{XX^T v} = \lambda \mathrm{v}}$

- No need to keep all eigenvectors – Retain only those with large eigenvalues, identified by gaps in the eigenvalue spectrum.

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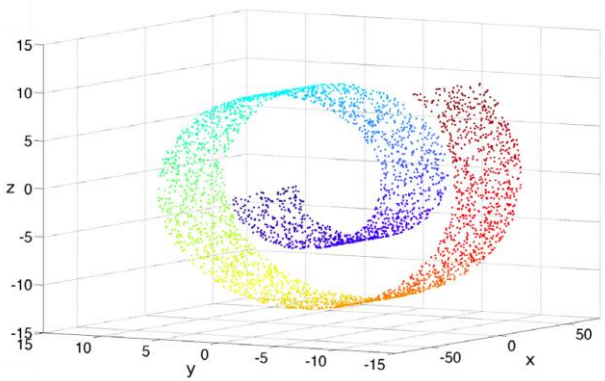
Non-linear Dimensionality Reduction: Diffusion Map

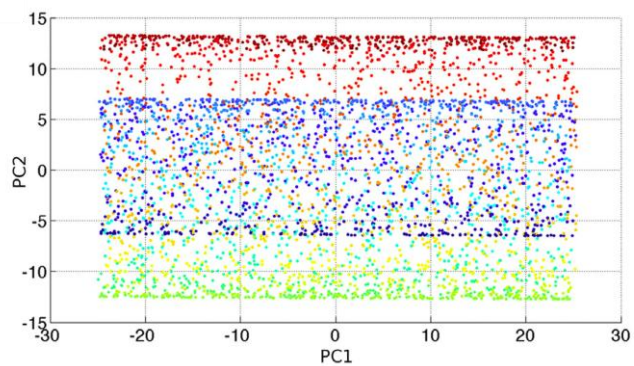
PCA offers a linear combination of input coordinates, which may be insufficient for describing complex datasets.

Non-linear dimensionality reduction methods, like Diffusion Maps, are more effective at capturing data with nonlinear coupling between degrees of freedom.



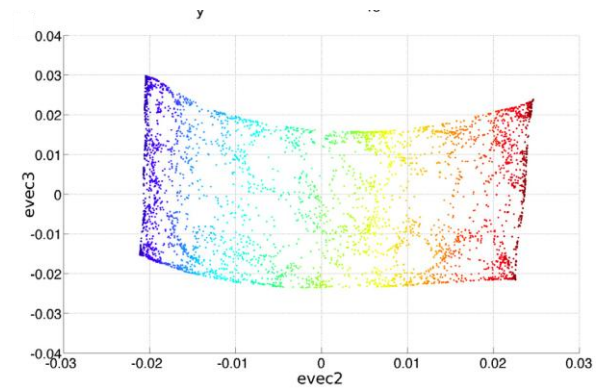
Swiss Roll

A rolled 2D surface in 3D space



PCA

Eigen-decomposition of the pairwise distance matrix (d_{ij}).



Diffusion Map

Eigen-decomposition of the nonlinearly scaled pairwise distance matrix:

$$A_{ij} = \exp(-\frac{d_{ij}^2}{2\epsilon}).$$

Chem. Phys. Lett. 509:1, (2011)

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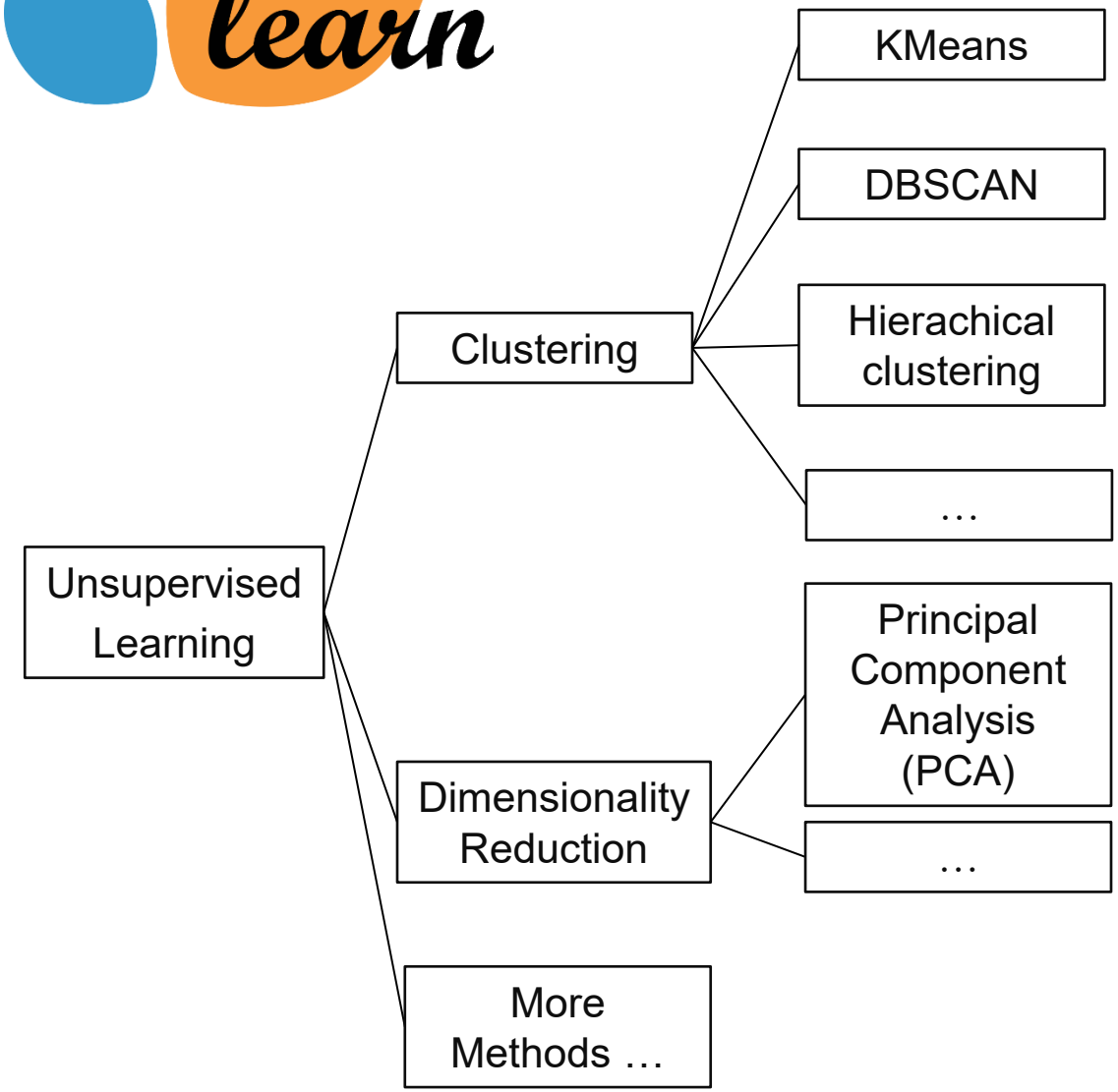
Scikit Learn




```
from sklearn.cluster \ import KMeans
```

```
from sklearn.cluster \ import DBSCAN
```

```
from sklearn.cluster \ import AgglomerativeClustering
```

```
from sklearn.decomposition \ import PCA
```

Scikit Learn provides efficient and user-friendly tools for machine learning in Python!

https://scikit-learn.org/

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